

SYMBOLIC POLITICAL VIOLENCE AND THE EFFECTS OF INSTITUTIONS ON PERCEPTION OF RISK (137 PP.)

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Memes have become commonly used symbols in the production of political violence. This study used politically violent memes as stimuli to measure changes in risk perception. Kent State University students ($n=133$) completed a survey which included meme viewing and risk perception ratings. The study was designed to measure reactions to viewing violent memes and examine the relationship between symbolic political violence and the role of institutions in risk perception.

This study created two new measures: a Perception of Risk Index (PRI) and a Symbolic Political Violence Index (SPVI). To test these constructs, participants viewed violent image-macro memes for a period of 60 seconds and recorded their reactions. Each meme was published online by well-known public institutions (Federal/National Government, Non-Profit/Non-Governmental, Academic, and Commercial Business). Participants were randomly divided into two equal groups: the treatment group viewed each meme with the caption of its original publishing institution, and the non-treatment group viewed each meme with a simple retrieval date.

The data show that the PRI and SPVI measures were highly correlated ($r = .52-.80$, $p < .001$). The also data show that there were no significant differences when memes were shown with an institutional caption versus memes shown with no caption ($p = .32$). Instead, the most accurate predictor for risk perception was political ideology. Self-reported conservatives scored lower PRI ($\beta = -0.90$, $p < .005$) and SPVI ($\beta = -0.88$, $p < .001$) compared to independents and

liberals. These findings have implications for how researchers and the public view memes, symbolic political violence, and risk perception, showing that each are complex and interdependent features of society.

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PERCEPTION OF RISK

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by

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TABLE OF CONTENTS

LIST OF FIGURES	viii
LIST OF TABLES	ix
ACKNOWLEDGMENTS	x
CHAPTER 1: INTRODUCTION	1
Scholarly Aims.....	1
Inquiry Background	2
Research Questions.....	3
Chapter Roadmap.....	4
Conclusion	6
CHAPTER 2: LITERATURE REVIEW	7
Risk	8
Risk Perception	9
Theories of Risk Perception.....	10
Affect Heuristic.....	10
Cultural Theory of Risk	12
Other Theories Related to Risk Perception.....	13
Risk Amplification.....	14
Social Amplification of Risk	15
Symbolic Political Violence	16
Symbols	17
Meme and Memetics.....	17
Online Memes & Image Macro	18
Memes and Social Change.....	19
Symbolic and Political Violence.....	20
Institutions	20
Institutional Influence	21
Institutional Trust.....	22
Theories of Institutional Trust.....	22
Conclusion	24
CHAPTER 3: METHODS.....	25
Institutional Review Board Approval	26

Subjects	26
Survey	27
Experimental Operationalization	27
Memes.....	28
Dependent Variable: Perception of Risk Index.....	34
Predictor Variables.....	35
Subject Variables	35
Additional Predictor Variables.....	36
Analysis	39
Ethical Considerations	41
Conclusion	43
CHAPTER 4: RESULTS.....	44
Participant Demographics.....	45
Participant Political Characteristics	46
Perception of Risk Index (PRI) and Symbolic Political Violence Index (SPVI) Characteristics	49
Meme Manipulation Check.....	50
Testing Hypothesis A: Correlation of PRI and SPVI.....	53
Testing Hypothesis B: Social Media Use Predicting Perception of Risk	54
Testing Hypothesis C: Group Difference by Caption.....	56
Testing Hypothesis D: Comparison of Institutions.....	58
Exploratory Analyses.....	59
Key Takeaways.....	62
Conclusion	62
CHAPTER 5: DISCUSSION.....	64
Summary of Findings.....	64
Key Findings.....	65
Key Finding: No Effect of Institutions	65
Key Finding: Political Ideology.....	67
General Discussion	68
Connections to Existing Knowledge.....	68
Broader Field Considerations	70
Advancements and Contributions	71
Limitations	72
Conclusion	75

CHAPTER 6: CONCLUSION	76
Introduction.....	76
Returning to the Original Inquiry Background	76
Potential Strategies to Mitigate Symbolic Political Violence	77
Look to the Future.....	78
REFERENCES	80
APPENDIX.....	96
Appendix 1.1: Flyer Recruitment.....	96
Appendix 1.2: Email Recruitment	97
Appendix 2.1: Survey Flow	99
Appendix 3.1: Survey Questionnaire	101

LIST OF FIGURES

Figure 1: WBS Meme	29
Figure 2: MCV Meme.....	30
Figure 3: ALS Meme	31
Figure 4: MON Meme	32
Figure 5: Symbolic Political Violence Index Item 1 Rating of Memes.....	51
Figure 6: Scatterplots of Correlations between Perception of Risk Index and Symbolic Political Violence Index across Memes with Fit Line	53
Figure 7: Graph Showing Influential Observations in OLS Regression.....	55
Figure 8: Mixed Effects Model of Main Effect of Each Meme on Perception of Risk Index, and Mixed Effects Model of Main Effect of Each Meme on Symbolic Political Violence Index	58
Figure 9: Perception of Risk by Grouped Political Ideology.....	60

LIST OF TABLES

Table 1: Perception of Risk Index (PRI) Items.....	41
Table 2: Symbolic Political Violence (SPVI) Items	42
Table 3: Subject Variables	42
Table 4: Additional Predictor Variables	43
Table 5: Participant Demographic and Background Characteristics	51
Table 6: Participant Background Political Characteristics	52
Table 7: Social Media Usage	55
Table 8: Internal Consistency of the PRI and SPVI	56
Table 9: Itemized PRI Per Meme, With Percent Change Between Groups.....	58

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CHAPTER 1: INTRODUCTION

Much of the meme culture in the United States is violent. Political mass murderers, assassins, and violent fanaticists use memes to inspire, convey, and spread their messages (Robins-Early, 2025). It is no longer a question of whether internet and online communications are part of the social cycle of violence. We are now living with the consequences of a violent political culture agglomerating with a violent online culture.

This dissertation looks at three main concepts: symbolic political violence, institutions, and perception of risk. The study aims to link these concepts through quantitative research methods and a positivist approach. The dissertation links these concepts to the growing research aimed at demystifying and fortifying the understanding of online symbols, cultures, and violence.

Scholarly Aims

The goal of this research was to create an original project that investigated human interaction with violent memes. The foundational theories and scientific approach were rooted in existing studies on risk, social communication, and political violence. The study had five points of unique contribution to academic research detailing the relationship between people, symbolic political violence, institutions, and risk. First, this study identified and estimated risk perception associated with symbolic political violence. Second, the study was the first to test amplification and attenuation of risk using symbolic political violence. Third, the study tested different institutions, which allowed for greater clarity on the effect of institutions on perception of risk.

The institutions were: the United States federal government; a nongovernmental organization; an academic journal; and a commercial website. Fourth, the study estimated the effect of symbolic political violence by using found artifacts (internet memes) developed in online spaces. Fifth and finally, the study used entirely symbolic indicators like cartoon memes to induce subjects' risk perception about the real world offline.

Inquiry Background

The inspiration for this concept was borne out of lived experience online and offline. During field work as a political organizer, I observed a common sentiment: *politics are getting more violent*, sui generis. This observation spanned my work at the Democratic National Committee in Community Outreach, the Ohio Democratic Party as a field organizer across rural and university populations in Ohio, and across eight U.S states as a canvassing director and administrator for non-profit and commercial firms. In this work, I personally contacted tens of thousands of voters, constituents, and community members, while my teams contacted hundreds of thousands. I observed this common sentiment broadly across political affiliations, social and religious sects, and geography. People were united in their blame: online social media.

Given the chance to explore this effect using cross-sectional data, I used the American National Election Studies 2020 Time Series dataset. This is a biannual survey distributed to 8,280 eligible United States voters. I examined responses to a survey question on violence perception: whether political violence had increased in the past four years (2016 to 2020). This was my attempt at operationalizing my field observation that there was increased perception of violence. I found that people generally agreed that violence had increased. I expected to find that people who had more engagement with social media and news would report more perceived violence. What I found was that users of social media site Reddit had significantly greater rates

of increased perception of violence (O'Brien, 2023). Moreover, people who viewed more online and television news media showed increased perception of violence. Something on these sites and in news media was associated with an amplified perception of violence. The theory which best described this effect is found in the Social Amplification of Risk Framework (Kasperson et al., 1988). This theory posits that institutions like news media or online communities can act as amplification stations, which can amplify perception of risk. From here, the next step was to test the theory more deliberately with experimental methods.

As a political scientist using a psychometric theory, the positivist paradigm is the most appropriate environment for this research. Using empirical analysis, my goal is to find and measure the effect of amplification. The inquiry was causal and deductive, requiring quantitative methodology. To quantitatively measure the psychometric properties of social amplification, I consider a survey instrument to be an adequate methodological choice. Thus, the study was developed as a response to a need: to understand the nature of risk and political violence and shed light on its social mechanisms.

Research Questions

This project was guided by three research questions. First, how does viewing symbolic political violence in memes influence perception of risk? Second, if a meme is transmitted by a trusted institution, does this change perception of risk? Third, to what extent does social media usage and thinking about political violence influence perception of risk from viewing symbolic political violence in memes?

Chapter Roadmap

This dissertation follows an organizational structure based on the chronological ordering of the study. The study itself was developed along this chronological path, beginning with observation and research, and culminating in a material experiment and subsequent analysis.

Chapter 1: Introduction provides a roadmap and background information for the dissertation study. First, it lays out each chapter in detail, as well as setting up the scholarly aims, and the goals for the research outcomes. The introduction also offers the background context through which the dissertation study was formulated. Following this, the research questions and the significance of the research questions are presented. This introduction concludes with a look to the literature to provide the conceptual framework and boundaries of the inquiry.

Chapter 2: Literature Review discusses the major foundations for the study and associated concepts. This review summarizes the existing literature which provided guidance, data, information, and theories that underpin the study. The chapter begins with a review of the literature on risk. The concept of risk has been explored in psychology, sociology, and political science. This includes an investigation of risk perception as a psychometric method, and what leads to risk amplification. The chapter continues with an introduction to the concepts of symbols, symbolic meaning, and visual semiotics. This leads to an exploration of memes, internet memes, and internet communication at large. Finally, this chapter concludes with a discussion of institutions, the influence of institutions, and how people interact with institutional trust.

Chapter 3: Methods describes the methods that were used to evaluate the study aims and hypotheses. This chapter begins with the basic construction of the project. Following the foundations of the project, participant recruitment and selection criteria are discussed. With the

participants selected, the chapter describes the survey instrument itself. This begins with detailing each study concept and its operational definition. Each variable is described along with their respective scoring schemes. The memes are identified and presented as they exist in the experiment. The main hypotheses of the study are presented and explained. After each element of the survey instrument is shown, the plan for statistical analysis is detailed, and hypothesis testing strategy is planned.

Chapter 4: Results presents the results of the survey. This chapter offers the totality of data collected by the survey. The chapter begins with a description of the participant pool, exploring the distributions in demographic characteristics. Each hypothesis is presented statistically (correlation, multivariate regression, t-test, ANOVA, and mixed effects regression), along with postestimation verifications to ensure validity. This focuses on the analyses which participants employed to describe each meme, highlighting key responses.

Chapter 5: Discussion focuses on interpreting the findings which were presented in the results chapter. Each hypothesis finding is explored in greater detail, featuring an autopsy of the unsupported hypothesis. After the findings are interpreted, the significance and contributions of these findings are presented. The chapter also addresses the principal limitations of the study, how it could be improved, replicated, and reformed to produce stronger results.

Chapter 6: Conclusion summarizes the dissertation contents, and connects it to real world application. The chapter uses the findings to make a case for the study's significance. This includes a look to future research which uses this study design. The chapter also focuses on recommendations to the field of political science, and how the subfields can adopt components from this study to achieve greater coherence. It concludes with a look to the future of academic inquiry in light of symbolic violence and politics.

Conclusion

As the political world continues to adapt to its environment, so too must its students. This dissertation is aimed at understanding our social world. The project serves to examine one part of a vast complex social web and derive meaning. From conception to conclusion, gaining a better understanding of the reality of symbolic political violence and risk perception drives this study. It is grounded in the experiences of especially young people online who are more prone to be exposed to and engaged in violence. Through this study, and hopefully many more in the future, we can begin to understand the relationship of memes and violence. Through coordinated efforts from governments and publics, community leaders and academic researchers, violence and peaceful solutions must be studied. This dissertation aims to contribute to this body of work, with the urgency and clarity of a mission: understanding violence as it exists in the social world, how people react, and what can be done in response.

CHAPTER 2: LITERATURE REVIEW

The introductory chapter presented the dissertation and its inspiration. This chapter will review the relevant literature. The ideas in this dissertation are from three key areas: risk perception, symbolic political violence, and institutions. This chapter provides a background by reviewing literature in these three areas.

First, the chapter discusses risk and risk perception. A review of theories of risk perception is introduced, including affect heuristic, cultural theory, and terror management theory. These theories represent different ways of thinking about how people experience hazards in the world. To complete the portion on risk, the chapter discusses amplification and attenuation of risk perception.

Next, this literature review details symbols and online memes. After the main definitions, the discussion turns to the transformative quality of memes as they interact with society. Finally, the symbols and political violence are connected to describe symbolic political violence.

The final portion of this chapter introduces institutions and trust. To begin, institutions are described as centers of social power and influence. This makes institutions important for creating meaning in society. However, institutions are constrained by how well they perform and whether they are trusted. This makes institutional performance and trust important in understanding how people view society. This portion discusses major developments in studies on institutional trust including specific linkages to risk perception.

Together, these concepts align to shed the greatest light on the dissertation. Risk, symbolic political violence, and institutions are at the heart of this study, and are therefore thoroughly reviewed to provide a grounded backing of the present study.

Risk

Risk is generally viewed as the likelihood of a hazard, danger, or loss. It can represent a broad range of social life, from chance of death to exposure to financial loss. Calculating risks, therefore, is a major part of political decision-making. Risk has been formally studied in science since the late 19th century, with its origins in mathematics (Mishura and Ragulina, 2016). Early models of risk focused on business outcomes, where probability of a business ending in ruin gave it the moniker ‘ruin theory.’ This mathematical view of risk is considered ‘traditional risk,’ ‘classical risk model,’ or the ‘Cramer-Lundberg approximation’ (Philipson, 1968; Mishura and Ragulina, 2016). This mathematical model continues to be used in contemporary business and actuarial settings. In the last century, the conception of risk has been developed further in social science scholarship. This has led to risk models being developed and applied broadly, across many disciplines. The current state of risk scholarship is robust, but in a state of limited definitional consensus.

Risk is studied in different ways. Political risk analysis evaluates states, commercial firms, and state-society relations (Korbin, 1979). These analyses result in the analysis of historical events and political environments geared at diagnosing and predicting potential losses. Medical risk analyzes factors leading to morbidity and transmission, which defines risk as a statistical association with a clinical outcome (Brotman et al., 2005). Social science studies have tried to identify and estimate the effects of risk on individual decision-making.

Risk Perception

A major development in risk research began with what is called the psychometric paradigm (Slovic et al., 1982; Slovic, 1987). In this measurement approach, risk became a matter of objective or subjective perception based on two factors: probability and magnitude (Rehani, 2015). This approach built upon previous studies by adding two dimensions beyond probability and magnitude: known-unknown and controllable-uncontrollable. These new dimensions allowed researchers to identify social institutions of risk, such as public concern and social trust (Slovic et al., 1986, Slovic, 1987; Kasperson et al., 1988, Renn et al., 1992; Earle & Cvetkovich, 1993; Kasperson et al., 2022). Risk perception has been described as related to social trust and locus of control, which can specifically be influenced by institutions (Riechard and Peterson, 1998; Breakwell, 2007).

Slovic et al. (1982) explained that risk perception is, “what people mean when they say that something is (or is not) ‘risky’” (p. 83). In other words, risk perception is a personal experience. However, risk perception is tied to real-world risk. Kasperson et al. (1988) stated that investigating risk, “is at once a scientific activity and an expression of culture” (p. 177).

Conceptually, risk perception can be described as an ontological relationship between objective and subjective (Fellenor et al., 2020). Events occur at an unknown objective magnitude and probability. Knowledge and control about these events also occur at an objective level. Public and personal response to risk is a subjective estimation based on the perception of the risk event. In Douglas and Wildavsky (1982), risk perception is described as, “a joint product of knowledge about the future and consent about the most desired prospects” (p.5). In this model, societies produce their own view of the natural risk environment. Changes to risk perception require changes to society and vice-versa. This position portrays risk perception as the outcome

of a complex historical pattern of social change. These views of risk through perception show that it is an intersubjective social process.

Risk perception research has now been used to investigate public responses to health and safety hazards (Slovic, 1987; Kaspersen et al, 1988; 1992; 2022). Social risk perception is often studied as a relationship between public perception and consequences of events (Lichtenstein et al., 1978; Fischhoff, Watson, and Hope, 1984, Kaspersen et al, 1988).

Theories of Risk Perception

What researchers try to determine is what people think about the world and, more importantly, why. The underlying origins of risk perception are subject to debate. There are many approaches to observing and studying psychometric risk perception. Some studied risks ranged from violent crime to vaccines (Flynn et al., 1994); street drugs to asteroids (Slovic, 2010).

There are a few ways risk perception researchers have approached the *why* of risk perception outcomes. One important way risk perception has been understood is through an Affect Heuristic. Another meaningful understanding of risk perception is found in the Cultural Theory of Risk. There are still many schools of thought and approaches to the study of risk perception outside of these main camps. The following details these two main theories and summarizes additional relevant theories.

Affect Heuristic

The relationship between an individual and their perception of risk is complex. How individuals make judgements on probabilistic matters is understood as behavior based on heuristics (Tversky and Kahneman, 1974). By using these mental shortcuts, researchers believe

that people use subjective judgement rather than objective probability to make assessment. One key finding which identified the peculiarities of this relationship is in the inverse relationship between risk and benefit.

Chauncey Starr's (1969) foundational study focused on how risk perception is associated to benefit perception. He argued that as perception of risk increases, perception of benefit decreases. The study found that public perception of risk is highly associated with voluntary participation. He found that the public is willing to accept voluntary risk "roughly 1000 times greater" than involuntary risk (p. 1237). The conceptualization of what risks are acceptable has become a necessary component in determining the impact of an activity. In this, it was not the nature of the risk activity, but the environment in which the risk activity exists in the mind of an individual. Importantly, the study found that acceptable risk was inversely related to the number of people participating in a risk activity.

This inverse-risk-benefit finding gained traction as a focal point of risk research. It has been found that risk and benefit may be positively correlated in reality, but negatively correlated in a person's perception (Alhakami and Slovic, 1994; McDaniel et al., 1995; Slovic, 2000). Alhakami and Slovic (1994) responded to this peculiarity with a model which explains the positive effect of benefit with its inverse, the negative effect of risk. The study attempted to deconstruct the overall affective evaluation of a risk hazard from the perception of risk and benefit.

The affect heuristic in perception of risk model works in sequence. First, humans are geared to understand risk and benefit through a dyad model of affect: hope and fear (Mowrer, 1960). Individuals bring this hope-fear affect into the reception of new information. This

information stimulates affect to produce a risk-benefit response (determining how risky and how beneficial something may be) based in subjective experience, not objective risk probability (Bateman et al., 2007; Slovic, 2000). In effect, this leads to an inverse relationship between risk and benefit. Where perceived low risk should yield low benefit and high risk should yield high benefit, researchers see the inverse: perceived low benefit yields high risk, and high benefit yields low risk. This relationship is important as it forms the way people view and participate in risk activities such as violence or online symbolization of violence.

Cultural Theory of Risk

Douglas (1970) writes that one is constantly “deriving cosmology from control systems” (p. 60). Her research on risk hypothesized that people’s abstract ideas like risk and order are determined by the social structures around them. In this, she noted that some risks are entirely relative to cultures, where risk can be viewed as transgressions of social norms. This was further developed with the idea of risk itself being based entirely in cultural production (Douglas and Wildavsky, 1982). In the perception of risk, the dangers of the world exist only through the perspective of an individual, bound by a limited set of knowledge cultivated by their society.

Individuals, not cultures, perceive risks, have concerns about society, hold worldviews. Culture, however, provides a collectively held set of customs and meanings, many of which are internalized by the person, becoming part of personality and influencing transactions with the social and physical environment. (Dake, 1991, p. 78)

Though risk is generated culturally, it is yet under the psychometric paradigm and is typically measured individually. Risk is not perceived uniformly across identity groups. Even when assessing individuals with technical understandings of risk and its measurement, identity differences are apparent.

In United States studies, white men report risk perception at significantly lower levels than other race-gender cohorts. Moreover, males of all races reported lower risk than all females, as well as white people of all genders reporting lower risk than people of color of all genders. According to a study by Flynn et al. (1994), for all 25 environmental hazards proposed, white males risk perception ratings were lower than the means of the other three groups nonwhite male, white female, and nonwhite female. Additional studies have found the same white male effect (Kahan et al., 2007; Slovic, 2010).

Slovic (2010) notes that this occurs in studies due to a cohort of white males, estimated at 30 percent of their respective cohort, who judge risks to be extremely low. Slovic refers to this group as LRWM, or low-risk white males. Importantly, this group was characterized by higher levels of education, income, conservatism, trust in institutions, and anti-egalitarianism. As demonstrated in the following chapters, these demographic factors play an important role in the conceptual framework of this dissertation.

Other Theories Related to Risk Perception

There are yet many approaches to understanding perception of risk. Risk perception has been an important source for understanding human interaction with hazard. Models like self-perception (Bem, 1967) and the elaboration likelihood model of persuasion (Petty & Cacioppo, 1986) assess individual risk perception as a psychological construct. Risk can also be considered an individual or social process of making meaning. Terror management theory is based in the study of self-esteem and the awareness of death (Greenberg et al., 1986; Pyszczynski et al., 2015). According to this theory, the inevitability of death can be the predominant factor in worldview formation. Reminders of death, or mortality salience, has been shown across over two

hundred studies to have a moderate effect on human thought (Burke et al., 2010). Although these theories aim to understand risk, external influences like institutions can change risk perception.

Risk Amplification

Institutions carry messages about risk, which can alter perception of risk. Institutional risk amplification can be formally explained through several lenses. To begin, it is important to understand how institutional messaging works. Two important models of institutional messaging are called Indexing Hypothesis and Cascading Activation Model (Traynor, 2018). In the Indexing Hypothesis, Bennett (1990) found that press communication follows a top-down flow of information. In other words, political messages created by the state are channeled directly to the public from official sources. This is a framing effect by using the institutional prowess of the state to legitimate messages. In this study, Bennett found that media framing matched governmental views due to an effective monopolization of information channels. This is important to risk information, as powerful institutions can shape and alter messaging. In an experimental setting, it has been found that institutions can carry complex signals about risk, but are often not isomorphic with the intent of the institution (Wahrman, 1963).

The Cascading Activation model (Entman, 2003) describes how communication can flow from political bodies to the public via intermediaries. Entman found that political messaging from U.S executive administrations traveled with minimal interference to non-administration officials like elected officials, then to media brokers who created and solidified framing effects, then to the public. Each institution along the way is an information station, but displays the way in which the public interacts with framed information rather than the natural risk environment.

The institution in this study is the information station, and the receiver is the participant; the message being sent is an internet image macro meme. To amplify risk, institutions must

communicate risk to the individuals via public information channels. If risk amplification is occurring via public institutions, conflict can increase. Marcks and Pawelz (2022) found that people would create meaning or stories about threats they face, called narratives of imperilment. They found that these narratives are necessary to activate perception of collective self-defense using violence. As groups feel more uncertain or at-risk, they may use violence as a means for what they believe is risk reduction.

Symbolic political violence may also cultivate a culture of violence. It has been found that media and mass communications can create a bias in people. This effect has been called “mean world syndrome,” when exposed to crime in media (Gerbner, 1998) or “scary world syndrome” (Andersen et al., 2024), when exposed to societal risks broadly.

This idea of cultivation, rooted in cultivation theory (Gerbner and Gross, 1976; Shanahan and Morgan, 1999), indicates that media contributes to perceptions of social reality whether the participant is aware or not. Boyd (2010) described an interaction called “context collapse,” where a message meant for one audience finds its way to another audience, causing a negative reaction. These findings, linked with this investigation, point to possible effects of risk amplification.

Social Amplification of Risk

The Social Amplification of Risk Framework (Kasperson et al., 1988) describes the social process of risk amplification. It describes risk amplification chronologically from event to effect. First, an event takes place. Next, that information is communicated between a transmitter and receiver. This stage occurs in loops described as feedback and iteration. After risk event and

risk communication, impacts are measured as outcomes to society. These can be widespread and interconnected in a manner described as ripple effects.

The Social Amplification of Risk Framework was designed to explain the social processes that influence risk perception (Kasperson et al, 1988; 2022; Renn et al., 1992). The process used in the model accounts for risk events, risk probability and magnitude, the social aspect of amplification. It identifies specific information channels which act as amplification or attenuation stations, and how they interact to influence risk perception.

Kasperson et al. describe an additional layer to the model: feedback and iteration. They argue that after risk events happen and are transmitted by information stations, there begins a secondary cycle of information flow between information stations. When messages are retransmitted through this feedback and iteration loop, amplification or attenuation continues. Importantly, these amplification and attenuation stations interact with themselves, each other, and public perception (Renn et al, 1992). These secondary impacts allow the model to describe risk perception at individual, institutional, and societal levels. In this, the messages themselves are what make the connection between a risk event, institutions, and the public.

This Social Amplification of Risk model has been applied to new risk events like global pandemics and social media. The research study proposed aims to examine a key portion of the model: the information station and its signal. Political violence transmission via symbolic imagery will be used to evaluate the influence of these processes as described in the model.

Symbolic Political Violence

Symbolic political violence is made up of two components: political violence and symbols. In semiotics, a sign is the most basic unit of information in social communication

(Peirce, 1867; De Saussure 1916; Burks, 1949; Scandura, 1970). A sign can be anything from a word to a physical object. Alone, a sign does not reveal any cultural context. An act of violence as a sign is an objective event; what the act of violence means is subjective and symbolic. When a sign adopts a meaning or cultural context, it becomes a symbol.

Symbols

Symbols are signs associated with a given meaning or context (Chick, 1999; Schirch, 2005). Symbols are important to risk amplification researchers because they are used as a means of risk communication. People and institutions use symbols to interpret the world. Symbolic interaction is described as the combination of interpretation of symbolic or non-symbolic acts or gestures, combined with definition to signal to the interactor how the intended message was received (Blumer, 1969).

For this study, perception of risk is the effect, and symbolic political violence is the medium through which this communication occurs. The receiver of information, sent from the information station transmitter is the individual. The transmitters, in SARF and in this study, are institutions.

Meme and Memetics

Baldwin (1896; 1909) postulated that biological concepts like natural selection and heredity relate to all aspects of humanity including social life. In digital or online spaces, communicative symbols often take the form of a meme. Since communications scholarship is much older than the internet, there must be a brief history of the idea of memes.

Early researchers developed the transmission model of communication, an early mathematics approach to the study of communication (Shannon and Weaver, 1949). A meme is

typically defined as a cultural unit that is transmitted amidst human populations in a biological fashion like a virus (Dawkins, 1976; Blackmore, 1999; Pech, 2003). Dawkins wrote that the meme parasitizes the brain, “turning it into a vehicle for the meme’s propagation in just the way that a virus may parasitize the genetic mechanism of a host cell” (p. 142). The way the meme spreads is through imitation or replication (Blackmore, 1999; Atran, 2001; Aunger, 2002).

The cultural unit is a symbol plus a social interaction, typically considered as a history or context (Chick, 1999; Wang & Wang, 2015). The infosphere is invisible relative to the biosphere, and thus relies on human interaction with culture to express (Gleick, 2011). Information is transmitted via users of media. This transfer is understood in the field of memetics, which was spawned from Dawkins’ term ‘meme.’ Memetics is similar to sociobiology, but has largely fallen wayside to internet study of memes as media objects (Chvaja, 2020).

Online Memes & Image Macro

Online/digital memes often employ an image or video plus text, which creates a new meaning to the original media object (Lankshear and Knobel, 2019). This image plus text combination is called an image macro. Paek (2025) describes the image macro meme as a first order semiotic text. There is an image, which Paek said is digested first, then text. In addition, there is a meaning of the original image macro meme from which other memes are derivative. Paek wrote on canonized image macro memes (CIMM): “a CIMM’s eventually canonized meaning will become so codified that the original image of a CIMM will become shorthand for a complex emotional landscape; an understanding of each CIMM’s highly specific emotional charge will be largely automated for a literate meme user” (p. 67). In other words, the semiotic

expression of the image macro meme is the image, the text, and the meaning landscape which connotes the original meaning.

In a process called condensation, Edwards (2022) notes that memes “allow users to condense complex ideas into simple packages that are honed through competition and reaction to achieve maximal effect and facilitate ‘shared cultural experience among users’ (30). For the purpose of this dissertation, the term meme takes the material internet conceptualization as well as the cultural informatics meaning together.

Memes and Social Change

Scholars have argued that the spread of online memes specifically invokes a transformation of information (Marino, 2015; Wiggins & Bowers, 2015; Wiggins, 2019). Each meme, then, becomes a cultural artifact that is constantly updated by the context of its most recent sender. In this way, online users become transmitters but also participants in the creation of social meaning. The process by which users impart their unique imprint unto a meme is described in the theory of heterogeneous volition. Howard (2017) places heterogeneous volition at the crux of creation, variation, and agency. He wrote that these digital communications are, “better understood as events instead of media objects” (p.119).

Memes and online communications also reflect the cyclical pattern of cultural development. Impression management theory (Goffman, 1959) declares that the self is a well-curated performance dictated by social situations. In internet-mediated communications like memes, the social element is constitutional to the production of self (Rettie, 2009; McEwan and Mease, 2013). Online identity is even shown to be a more compressed version of its offline self. McEwan and Mease (2013) found that identities not only bleed between offline and online, but that the online identities tend to be ‘crystallized’ versions of the offline counterpart. Through

this, we can see that the socio-technical institutional online space has material effects on the content and quality of social communication between people.

Symbolic and Political Violence

Political violence can be direct, indirect, physical, structural or cultural. Symbolic violence, however, is exclusively indirect. Galtung (1990) stated, “[b]y ‘cultural violence’ we mean those aspects of culture, the symbolic sphere of our existence ... that can be used to justify or legitimize direct or structural violence” (p. 291). Della Porta (2016) wrote that “violent entrepreneurs” can “activate symbolic resources” in the production of political violence (p. 19). Thapar-Björkert, Samelius, and Sanghera (2016) relate this to power in that symbolic violence, “simultaneously legitimises and sustains other forms of violence” (p. 144).

Bourdieu (1999) declared that symbolic violence, unlike an act of violence, can only be communicated to a person predisposed to understanding it, whereas someone unaware would not receive the communication. This social nature of symbolic violence is what Recuero (2015) argued gives “super powers” to social media environments in their ability to amplify symbolic violence. These are important to militants and supporters to drum up support for violence as they require sociocultural infrastructure like imperatives for destruction, images of enemy evil, and narratives of a virtuous ingroup identity (Rothbart, 2019).

Institutions

This study focuses on institutions from the perspective of institutional trust. Douglas and Wildavsky (1982) wrote, “in risk perception, humans act less as individuals and more as social beings who have internalized social pressures and delegated their decision-making processes to institutions” (p. 80). Violence has been studied in its amplifying effects through institutions like

social media (Hill, 2001; Slovic, 2020) and television media (Tartaglia, Gattino, & Bimbo, 2019); as well as in public perception at large (Pidgeon et al., 2003).

Institutions are the transmitters of information because they are devices for decision-making. In the Social Amplification of Risk Framework model, signals are transmitted from the information station to the receiver. Institutions are the information station device in the real world. The information station itself can have symbolic meaning to a receiver (Kasperson et al., 1988). That is to say that the same message from different information stations may convey different meanings to different receivers. Different institutions, as information stations, convey varying degrees of risk amplification. Recalling that risk perception requires a societal production (Douglas and Wildavsky, 1982), institutions play the key role in developing and sustaining knowledge and consent about risk. Institutions can produce consent via coercion or trust.

Institutional Influence

Institutions have been shown to have effects on perceptions of risk (Kasperson et al., 1988; Slovic, 1993; Frewer et al., 2003; Dai et al., 2022). Kasperson et al. (1988) identified key amplification stations in risk communication as scientists, news media, activist social organizations, opinion leaders, personal networks, and public agencies. They noted that public responses to risk events tended not to focus on the institutional environment of the risk signals.

Frewer et al. (2003) found that the type of institution matters to public perceptions of risk. Their study concluded that the public and scientific institutions view risk communication differently. They found that the way institutions communicated, along with the type of institution, impacts public response to risk signals. According to Quéma (2015), the ability for an institution to actually materialize their influence is done by symbolic power, using signs and

signals. They do this by generating messages, which can bolster their own legitimacy.

Institutions like government regulatory bodies, nongovernmental organizations, and commercial firms can change the level of risk perceived by individuals *ceteris paribus*.

Institutional Trust

Researchers into institutions found trust to be an integral part of institutional influence. They developed a system of measuring trust that people have in certain institutions, as well as their trust in others. In this model, there are two forms of trust: social trust and institutional trust (Daskalopoulou, 2019). Social trust is based on perception between people, and institutional trust is perceived by a person about an institution. Schilke, Reimann, and Cook (2021) described two important strains of social trust: particularized trust and generalized trust. Particularized trust is narrow and personal, whereas generalized trust is impersonal and social. These are important areas to consider in institutional research, as trust necessarily varies from person to person. In social trust, the ability or desire to trust is measured by a perception of a generalized ‘other’ (Robinson, 2001; Sønderskov and Dinesen, 2016; Daskalopoulou, 2019). Institutional trust is different from these two in that it relies on an individual perception of a political entity and its cultural environment.

Theories of Institutional Trust

“[A]ll communication involves risk. This risk is a very important morphogenetic factor because it leads to the establishment of institutions that guarantee acceptability even in the case of improbable communications” (Luhmann, 1992, p. 255). Institutions act as loci of trust. There are three main categories of institutional trust theories most useful for this investigation. The first category is socialization-developmental theories, the second is cultural, and the third is communicative.

In developmental models of institutional trust, cognitive affect serves an important purpose in building and maintaining trust in social institutions. From early childhood onward, people engage in social processing of the political world, which gleans norms, values, and ideas about institutions (Greenstein, 1960; Easton and Hess, 1962; Bender, 1967; Easton 1968). Socialization is considered a developmental social process through which individuals understand the world.

In development, there is some implicit conferral of trust from parents that is considered basic personality (Kardiner, 1949; Bender, 1967). In this conception, there is a base level of trust that informs all social processing. When it comes to political systems, however, this must be learned. The social process of understanding institutions comes from explicit and implicit political socialization. The first formal political-social processing that a child encounters is the role of hierarchy in the family (Kardiner, 1949; Bender, 1967). Family hierarchy then reflects society and teaches children through development to understand authority. In this conception, conferred power hierarchically is an implicit socialization process through which politics are developed.

Bender (1967) wrote that political socialization can occur implicitly first, but continues explicitly through cultural values, social systems like hierarchicalism, and accompanying institutions like legal systems. Even with these distinctions, Daskalopoulou (2019) found that these socializations- and even social and institutional trust themselves- are considerably interrelated.

Dai et al. (2022) studied perception of COVID-19 risk and reported that as people received more information from risk management institutions, their trust in the institution

increased as well as their perception of risk. A takeaway from these studies is that institutions have a powerful influence on risk perception and public response to risk events.

In this dissertation study and in regular daily life, the participant either knows or does not know the institution directly. The signal of each institution gains meaning through the perspective of the participant. Participants may know employees or specific work done by an institution (a colleague who works for the Federal Bureau of Investigation). They could also indirectly know the institution as a concept (generally understanding what the institution does, its goals, its history). These complex relationships are important to consider in developing an adequate and accurate depiction of risk, symbolic political violence, and institutions.

Conclusion

When someone sees a violent meme published by an institution, what happens? To understand this question, all of these various theories and studies must be considered. In reviewing the research, taking note of the methods and findings, a plan can be made to seek an answer. Risk, symbolic political violence, institutions, and many other important factors must be turned into measurable variables.

This chapter provided the necessary review of the relevant scholarship leading to this dissertation. Many of these concepts will return in the discussion chapter, where the results are tied back to these foundational works. The proceeding chapter, Methods, turns the concepts discussed in this chapter into an experimental design.

CHAPTER 3: METHODS

The previous chapter reviewed the relevant academic work which provided inspiration and foundational support for this investigation. This chapter will provide the mechanics of transforming these concepts into measurable phenomena. In order to do this, a study was devised in which a survey was distributed Kent State University students. This survey contained questions which lead to findings on the relationship of risk, symbolic political violence, and institutions.

Overall, this is a positivist study guided by a quantitative empirical approach. The goals were to create psychometric measures of viewing symbolic political violence on the perception of risk, and the role institutions play. The data was collected online and analyzed using a statistical software to test hypotheses. This chapter provides the comprehensive setup, operationalization, and execution of formulating propositions, testing them, and verifying them using empirical means.

This chapter will be guided by the following organization. First, a full explanation of the research model is presented. Second, the experimental stimulus, four image macro internet memes, are described in detail. Third, the measures and variables will be discussed and prepared for analysis. Fourth, analysis of the data is described. Finally, ethical considerations of this study are presented. This chapter will describe the mechanics of the dissertation study from conception to completion.

Institutional Review Board Approval

In order to survey this population, there were ethical and legal obligations to earn the ability to collect data. First, I applied to the Kent State University Institutional Review Board via the Office of Research Compliance. The study was approved for Level 2 Expedited Review on October 31st, 2024, and remains active until expiration on October 30th, 2029.

Subjects

The subjects in the study were Kent State University students who were registered for at least one course in the Spring 2025 semester. The recruitment was done using two tools: emails and flyers. The email recruitment was conducted by making a Request for Release of Student Information with the Office of the University Registrar. This data request allowed for the release of a list of 2,500 students currently enrolled at Kent State University. The number was limited to satisfying a 10% return rate on the email survey for 250 respondents. Flyers were also distributed at the Kent State University Main Campus in the student center, library, and five bus stops central to campus.

Participants could interface with the survey instrument via two methods. For email recruitment, they could click a link on desktop or mobile, and on flyers a scannable QR code could be accessed via mobile camera. Both direct participants to a unique url hosted by Qualtrics XM, a survey software. Here, the survey is conducted. The Qualtrics XM survey was sent via email to the random 2,500 students on three occasions: April 10th, April 17th, and May 1st, 2025. Flyers were distributed on April 10th.

Survey

The survey instrument is the tool used to observe, capture, and measure psychometric effects of this experiment. The survey consisted of 38 items: one Informed Consent question, seven sociodemographic questions (age, gender, nationality, race, education, employment, and political ideology), three questions assessing social media usage, engaging with political violence, social dominance orientation. After viewing memes, there were two indices assessing Perception of Risk (six items) and Symbolic Political Violence (five items). After the meme viewing indices, there were two questions about trust in institutions and confidence in institutions, placed after the meme assessments to avoid a priming effect. At the conclusion, there is a debrief with a text-entry for feedback to the researcher, followed by consent to enter the compensation raffle.

There were three stages of the survey: baseline survey completion, meme viewing, and post-meme survey completion. The first section collected some sociodemographic information as subject variables. The second stage was the meme-viewing portion and completion of indices assessing Perception of Risk and Symbolic Political Violence. The third stage contained additional sociodemographic questions and concluding items.

Experimental Operationalization

To test these data for the hypotheses, variables were created using both new and existing tools. Participants will answer questions using the survey tool Qualtrics. The dependent variable measures perception of risk, which is measured by index. The moderating variable is binary whether a professional institution is (1) or is not (0) mentioned in the survey. There are also

subject variables related to demographic characteristics, and predictor variable items which are expected to show positive association with the dependent variable.

In the second stage, the participant will be directed to a full page viewing of a single image macro meme. Each participant viewed all four memes. Their instruction was to view each image for 60 seconds before continuing with the survey. Page breaks were added between each meme and subsequent post-meme question items. To ensure an appropriate time was spent viewing the meme, participants were blocked from advancing until the viewing time was completed.

Participants were asked to answer questions from predictor and outcome variables after each period of 60 second meme viewing. Participants alternated between viewing the meme for 60 seconds and answering the same questions about the meme for a total of four repetitions. Upon completion, participants were then prompted to complete a voluntary form to enter a compensation raffle.

Memes

In order to measure perception of risk, a stimulus must be induced for participants. In the survey, the participants are asked all textual questions except one portion which displays memes. During this portion, participants are shown four image macro internet memes. Each meme was selected because they were already published by institutions publicly. The four memes are: 1) White Boy Summer; 2) McVey; 3) All Lives Splatter; and 4) Hoppe. They are sourced from the Anti-Defamation League Center on Extremism; Joint Counterterrorism Assessment Team; National Counterterrorism Center, Federal Bureau of Investigations, and Department of Homeland Security; and Know Your Meme.

The main materials external to the survey are the image macro memes selected for experimentation. This is a very standard format of internet meme. The four memes selected for this study are all image macro style memes. Each meme was selected after a careful review of literature and institutional publication. The four image macro memes are: 1) White Boy Summer (WBS) ; 2) McVey (MCV); 3) All Lives Splatter (ALS); and 4) Monarchy (MON). They are sourced from the Anti-Defamation League Center on Extremism; Joint Counterterrorism Assessment Team: National Counterterrorism Center, Federal Bureau of Investigations, and Department of Homeland Security; and Know Your Meme.

The first meme, ‘White Boy Summer,’ is a meme from 2021 that has been used by right wing extremists online in concert with violent messaging (Stewart et al., 2023; Balan, 2021; Waldek et al., 2021). The selected meme published by the Nevada Anti-Defamation League features a fashwave aesthetic, nazi symbology, and a caption, “Hitler approved wbs [white boy summer]” (Nevada ADL, 2021). The Anti-Defamation League is a non-profit, nongovernmental civil rights organization based in New York City, United States.

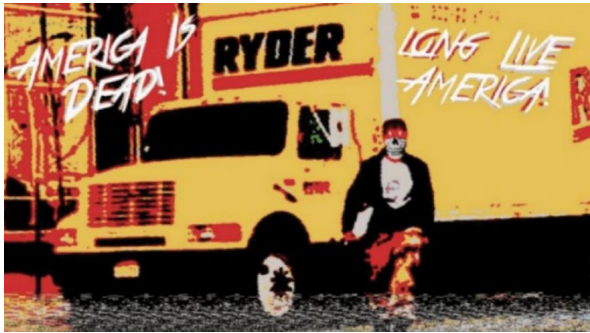
Figure 1
WBS Meme

Institution mentioned (1)	Institution not mentioned (0)
 Hitler approved WBS.	 Hitler approved WBS.

Source: ADL (2021). "White Boy Summer:" From Meme to Mobilization." Anti-Defamation League: Center on Extremism.	(Retrieved from the Internet 2/27/2024)
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The second meme, ‘McVey’ is a similar fashwave aesthetic displaying one person, Timothy McVey, the Ryder truck used in the Oklahoma City bombing, and the words, “Ameriga [sic] is dead,” and “Long Live Ameriga [sic]” (JCAT, 2022). This meme does not display an act of violence, but instead makes indirect reference to violence by showing a weaponized truck bomb. The institution and source is a multi-agency United States federal law enforcement publication produced by the Joint Counterterrorism Assessment Team. This team consists of the National Counterterrorism Center, the Department of Homeland Security, and the Federal Bureau of Investigation.

Figure 2
MCV Meme

Institution Mentioned (1)	Institution not mentioned (0)
	
Source: Joint Counterterrorism Assessment Team. (2022) "Use of Memes by Violent Extremists." National Counterterrorism Center, Department of Homeland Security, Federal Bureau of Investigation.	(Retrieved 3/17/2024)

The third meme, ‘All Lives Splatter’ is identified as a politically violent meme by Know Your Meme, the Southern Poverty Law Center (Ward, 2018) and by academic researchers (Miller and Hayward, 2019; Jenkins and Butterworth, 2020; Rozdilsky and Snowden, 2021; Savitzky and Ciddell, 2023). The meme is black and white, showing a cartoon car running over stick figure people with the text, “All Lives Splatter” across the top, and “Nobody cares about your protest” across the bottom. The institution and source that published this iteration was the *British Journal of Criminology*, an academic periodical published by Oxford University Press.

Figure 3
ALS Meme

Institution mentioned (1)	Institution not mentioned (0)
 The image is a black and white meme. At the top, the text "ALL LIVES SPLATTER" is written in a bold, sans-serif font. Below the text is a white silhouette of a car. The car is running over several stick figures. One stick figure is lying on the ground to the left of the car, another is being run over by the front wheel, and a third is running away from the car to the right. At the bottom of the image, the text "NOBODY CARES ABOUT YOUR PROTEST" is written in a bold, sans-serif font.	 This is an identical copy of the image in the first column, showing the "All Lives Splatter" meme with a car running over stick figures and the text "NOBODY CARES ABOUT YOUR PROTEST".
Source: Miller, V. and Hayward, K. J. (2019). 'I did my bit:' Terrorism, Tarde and the vehicle ramming attack as an imitative event. <i>The British Journal of Criminology</i>, 59(1), 1-23.	(Retrieved 2/27/2024)

The final meme, ‘Monarchy,’ was selected for its overt expression of symbolic political violence. The meme features two portions. The first shows a crowned Wojak, representing a

monarchist, and a Pepe the Frog personified as Hans-Hermann Hoppe, an economist famous for anarcho-capitalist anti-democratic writings, holding an assault rifle. The text reads, “Here is a list of reasons why monarchy is superior to democracy//1.king’s wealthfare [sic] is tied to the wealth of his subjects.” The second portion features the same monarchist Wojak, this time showing a shocked expression. Hoppe is displayed this time as a meme ‘psycho wojak’ holding a bloody shotgun toward the monarchist wojak. The text reads, “Here’s another it’s easier to behead a king than a whole parliament.” This meme was not published by a professional institution, but can be found on Know Your Meme, an internet meme repository database.

Figure 4
MON Meme

Institution mentioned (1)	Institution not mentioned (0)
	
Source: (March 2nd, 2023). Political Compass. Know Your Meme.	(Retrieved 4/12/2024)

Many researchers have pointed to Know Your Meme as a repository (Bratt and Smith, 2022; Rogers and Giorgi, 2023); a database (Shifman, 2013; Rogers, 2022); an archive (Țăran, 2018); a public log (Hanganu-Bresch, 2017); a reference site (Milner, 2018); and as “a well-

regarded tool for learning about the origins and meanings of popular memes” (Boyle, 2022, p. 89). There is criticism of relying on Know Your Meme as a sole authority in research due to being for-profit rather than user-generated nonprofit, and its ‘opaque’ methods and curation process (Pettis, 2022). Therefore, Know Your Meme is included to offer a range of institutions from professional institutions to informal public institutions. In this case, Know Your Meme is an informal public institution.

Together, these memes each meet criteria for displaying symbolic political violence. They are each sourced from public documents easily available online to mimic average daily use. Finally, these are memes that have been reproduced enough times to merit collection and description, which helps to maintain reliability of each measure. Large size memes can be found in the Appendix.

Measures and Variables

This study tested a total of 28 subject variables and four iterations of two repeated variables: one dependent variable and one predictor variable. The organization of the variables were based on presentation to the participants. First, there were seven demographic variables, eight social media variables, five online behavior variables, and four social dominance orientation variables. All were independent variables. Next, the participants viewed the stimulus memes and responded to the repeated measures. After completion of the stimulus meme viewing, participants were presented with four variables on institutional trust, and four variables on institutional performance. These variables were given last so that participants were not primed to think about institutions before viewing the memes and captions.

Dependent Variable: Perception of Risk Index

The outcome of this study, measured by the dependent variable, was perception of risk. An index was created to combine several elements of perception of risk into a composite score. This composite score is called the *Perception of Risk Index (PRI)*. All questions were coded with the same 1-7 Likert Scale. The scale shows 1=Strongly Disagree; 2=Disagree; 3=Somewhat disagree; 4=Neither agree nor disagree; 5=Somewhat agree; 6=Agree; and 7=Strongly Agree. The codes of six questions combined were summed, and PRI scores can range from a low of 5 to a high of 35, with one “no change” option for descriptive statistics.

The first item asks participants about personal risk perception, “This meme is a threat to my personal safety.” The second item asks about public risk perception, “This meme is a threat to public health and safety.” The third item asks about increasing violence, “This meme will cause an increase of violence toward the victims depicted in the meme.” The fourth item asks about real world violence, “This meme is likely to cause violence in real life.” The fifth item asks about attitude incitement, “This meme is likely to incite violent attitudes.” The final item asks about a potential non-effect, “This meme is unlikely to cause any change to violence or attitudes about violence.” This allows for participants to express an inverse viewpoint, and is coded with values reversed (1-7 becomes 7-1).

Table 1

Perception of Risk Index (PRI) Items

Survey Item Code	Survey Question Full Text
Personal	This meme is a threat to my personal safety
Public	This meme is a threat to public health and safety
Increase violence	This meme will cause an increase of violence toward the victims depicted in the meme

Real violence	This meme is likely to cause violence in real life
Incite	This meme is likely to incite violent attitudes
No change	This meme is unlikely to cause any change to violence or attitudes about violence

Note. The No Change item was reverse-coded prior to PRI scoring.

Predictor Variables

There were two sets of independent variables, or predictor variables. The survey included these predictor variables that were expected to have a positive relationship with the dependent variable (See Hypothesis A). This item is called the Symbolic Political Violence Index (SPVI). It served as a manipulation check to determine that the methodology was internally valid. Like the PRI, the values of all five items were summed, which produced a range of values 5-35.

After each meme, participants answered the PRI and SPVI questions one time. They answered this question block after the *Perception of Risk Index* for a total of four repetitions. All items were coded with answer options in the 1-7 Likert scale listed.

Table 2

Symbolic Political Violence (SPVI) Items

Survey Item Code	Survey Question Full Text
Act	There is an act of violence depicted in this meme
Real act	This meme is referencing real-world violent acts
Lethal	The violence depicted in this meme is lethal
Political	This meme clearly presents a political viewpoint
Pro-Violence	This meme clearly presents a pro-violence stance

Subject Variables

The research participants carry their unique characteristics into the experiment. In order to address these important differences, several questions are asked to gain a picture of each

participant. Table 3 shows the background demographic characteristics which were collected.

These act as predictor variables in some analyses, and help guide the findings toward the fullest picture of each participant.

Table 3
Subject Variables

Survey Item	Variable Type	Codes
Age	Continuous	18-99
Gender ID	Categorical	Select all that apply: Man, Woman, Non-Binary, Genderfluid, Transgender Woman, Transgender Man, Other, Prefer not to say
Nationality	Categorical	United States, Other[text entry]
Race/Ethnicity	Categorical	Select all that apply: White or Caucasian, Black or African American, American Indian/Native American or Alaska Native, Asian, Native Hawaiian or Other Pacific Islander, Other, Prefer not to say
Education level	Categorical	Some HS, HS diploma or GED, Trade or Vocational, Some college, Professional Cert., Associate, Bachelor, Master, Professional Deg., Doctoral
Employment	Categorical	Select all that apply: Full time, part time, unemployed, homemaker/stay at home, student, retired, other
Lib-Con	Ordinal	Very liberal, liberal, somewhat liberal, neither liberal nor conservative, somewhat conservative, conservative, very conservative

Additional Predictor Variables

There were additional predictor variables related to how people interact with risk and institutions. Each of these account for important components of a participant's characteristics prior to beginning the survey. Based on previous studies and existing knowledge, the following are the key participant characteristics selected to control and measure before conducting analysis.

Each of these serves as a predictor variable and to provide the necessary descriptive statistics to the survey pool.

The first category is social media usage. This item asks about each of the major social media platforms and how often they are used. The second category addresses how much people indirectly engage with political violence through their thoughts and behavior online. The third category addresses their social identity through social dominance orientation, or how much they prefer social dominance in society. Finally, there are two categories which ask participants about their trust in the institutions that were part of the study. The following table displays the full list of additional predictor variables used in the study.

Table 4
Additional Predictor Variables

Variable Category	Survey Items	Coding Scheme			
		Frequency	Agreement	Trust	Confidence
Social Media Usage	Facebook,	1) Never			
	X/Twitter,	2) Rarely			
	Instagram, TikTok, Reddit, WhatsApp, Snapchat, YouTube, Other	3) Occasionally			
Engaging with Political Violence	Think about political violence	4) Sometimes			
	Think about violence in general	5) Frequently			
		6) Usually			
	View memes depicting political views online	7) All the time			
	Discuss political violence on social media				

Variable Category	Survey Items	Coding Scheme			
		Frequency	Agreement	Trust	Confidence
Social Dominance Orientation	The majority group should get a say over the minority group		1) Strongly Disagree		
			2) Disagree		
			3) Somewhat disagree		
	We should not push for group equality		4) Neither agree nor disagree		
			5) Somewhat agree		
	In setting priorities, we must consider all groups		6) Agree		
			7) Strongly Agree		
Trust in Institutions	Choose how much you personally trust each of the institutions: Federal/National Government Organizations, Non-Profit/Non- Governmental Organizations, Academic Organizations, Commercial Businesses			1) Strongly do not trust	
				2) Do not trust	
				3) Somewhat do not trust	
				4) Neither trust nor do not trust	
				5) Somewhat trust	
				6) Trust	

Variable Category	Survey Items	Coding Scheme			
		Frequency	Agreement	Trust	Confidence
				7) Strongly Trust	
Confidence in Institutions	Choose how much confidence you have in each institution's ability to perform their organizational goals: Federal/National Government Organizations, Non-Profit/Non- Governmental Organizations, Academic Organizations, Commercial Businesses				1) Strongly not confident 2) Not confident 3) Somewhat not confident 4) Neither confident nor not confident 5) Somewhat confident 6) Confident 7) Strongly Confident

Note. The second and fourth items in Social Dominance Orientation were reverse-coded. Bold font was presented to the participants.

Analysis

After the collection of survey data for variable measurement, analysis is conducted. The program used for statistical analysis is STATA 18/19 BE. This is a purchasable statistical analysis software used to apply testing and treatment to raw data. The survey data is collected via

Qualtrics XM, which automatically checks data for validity. The software identified two instances of “straightlining,” or answering all questions with a similar repeated value. These data were eliminated from the dataset prior to export. The data generated by Qualtrics created a comma separated values (CSV) file. The data in this CSV file is raw, meaning it required cleaning in STATA to eliminate missing values and address any potential issues. Once the data was exported into a CSV file, it was uploaded directly to STATA for analysis.

The data was cleaned and prepared for analysis by removing all missing values, then renaming variables into legible codes. This means creating meaningful signifying codes to determine variable names when conducting tests. Finally, the cleaning process entailed recoding variables such as the non-effect item of PRI and Social Dominance Orientation which are reverse-coded. This simply means rearranging the values that were previously one thru seven to now be measured seven through one.

This study employs a 2 x 4 repeated measures ANOVA (two groups: institution mentioned/institution not mentioned x four memes). There are three required arrangements, however, as one is a between-group (institution mentioned/not mentioned) and one is within-group (among the four memes), and the interaction effect within-between. To test the main effects of the ANOVA for between-group at a power of .80, with an anticipated small effect size of $r = .20$, and $\alpha = .05$, a sample size of 207 was calculated by G*power 3.1 statistical analysis software (Faul et al. 2009).

Qualitative Text Responses

An important part of the survey analysis was the text-based responses asked of participants after viewing each meme. Upon completion of the meme viewing period of the

survey, participants were asked to describe the meme. The question stated, “How would you describe the defining characteristics of this meme? Please be as clear and specific as possible.”

This was repeated four times per participant and was the same for both caption groups.

This question was meant to collect further details from the quantitative analysis and provide insight into the participants’ ideas. The answers from some participants are selectively used to support quantitative findings in the discussion. This provides a supplementary but ultimately more complete depiction of the participant risk perception experience, in the participants’ own words. The text responses are presented as in vivo quotations, meaning responses are reported exactly and in full. This follows a model described in Charmaz (2014) model of in vivo coding, which is used to, “simultaneously reflect condensed meaning of a general term and reveal an individual’s fresh perspective” (p. 135). Additionally, some text responses are counted using keyword search to count simple frequency of word usage.

Ethical Considerations

There are two main ethical concerns for participants of this study. The first concern is that participants could have a negative psychological and physiological response to viewing violent memes. The second concern is that violent memes could influence the participants in real life toward violent action. The following section addresses the mitigation strategy against these hazards.

The first concern is about the immediate impact of viewing a violent meme. Jordan (2014) wrote that distribution of potentially harmful imagery is a problem in many social science research projects. They suggested that researchers must make deliberate efforts to notify the participants in the informed consent process. This study specifically notes in the informed

consent process: “There may be some questions or images (memes) which may make you feel uncomfortable. You will be shown four instances of memes displaying symbolic political violence, which some can find disturbing. If you do not wish to proceed, you may skip it and proceed to the next question or exit the survey. Your participation is expressly voluntary.” Additionally, email and flyer recruitment displays the title of the study, which informs participants that the study is about symbolic political violence.

The second concern is that viewing these memes might influence the participants toward violence. The memes used were created for distribution, not natural photographs of real violence. The memes act as calls to violence, which could be accepted by the participants. There are two levels of mitigation in this study. First, there is a detailed debrief for all participants at the end of the survey, even if they exit early. This describes how the memes were selected, why they were used, and how they will be analyzed. This provides necessary context that the memes were identified as extremist and to be used for scientific purposes. Second, the memes in the study have already been published and distributed publicly. Each of the memes were specifically chosen because they were pre-existing accessible media items. Institutions representing the federal government, a nonprofit organization, an academic journal, and a commercial website were the publishers. The public, including researchers, rely on these professional institutions’ adherence to codes of ethics in distributing this material (Keith et al., 2006).

These ethical strategies are made to minimize the hazard to each participant. This study was deemed Level 2 by the Kent State IRB and was treated with an appropriate level of care. The potential hazards to participants and society are serious and have been planned and addressed thoroughly in this study.

Conclusion

This study is focused on using internet memes to elicit perception of risk and measure differences. This chapter detailed the full experimental research design of the study. This included the participant population, the data collection, the analysis, and the materials used for measurement. In these explanations, each element of the study is operationalized from concept to measurable survey responses.

This type of study has not yet been done, requiring unique contributions. The memes used in the study have been online for years, but have not been part of a research study. As well, the study used two new measures: the Perception of Risk Index and the Symbolic Political Violence Index. The nature of this study is exploratory in some methods but is grounded in classic social science methodology. The next chapter, Results, focuses on the outcome of the study.

CHAPTER 4: RESULTS

This chapter presents the totality of findings from this research study. This chapter follows Chapter 3: Methodology, which explains the study design from planned analysis. This chapter presents all of the data from planned analyses to uncovered relationships. These data fully represent the findings from the research study and concludes the investigative portion of the study.

There are four main hypotheses in this study. First, Hypothesis A predicts that the two created constructs, Perception of Risk Index and Symbolic Political Violence Index, will share a positive relationship. Hypothesis B predicts a significant and positive relationship between Social Media Usage and Perception of Risk. Hypothesis C predicts that memes with an institutional caption will have significantly higher Perception of Risk scores than memes without institutional captions. Finally, Hypothesis D predicts that the non-professional institution, coded as Meme 4, the MON Meme, will have an attenuated effect on Perception of Risk relative to the professional institutions coded by Memes 1-3.

Participants and Study Completion

A total of 207 Kent State Students responded to the study recruitment announcement and opened the Qualtrics survey link. Qualtrics flagged two participants for straight lining their responses, and they were removed. Of the remaining, 22 participants exited the survey prior to viewing any memes, and 50 exited the survey prior to viewing all four memes and completing all of the survey measures. The total duration for the study required at least an 80% completion rate

to include all key variables minus the debrief page and raffle participation. Thus, 133 (64.3%) participants completed the survey fully. The study used these 133 participants for all analyses.

The first survey was collected on April 10th, 2025, and the final survey was collected on June 30th, 2025. Three participants (1.5%) accessed the survey via flyer QR code. The remaining participants accessed the survey via email recruitment link.

Participant Demographics

Table 5 shows a summary of participant demographic data. The mean participant age was 28.92 years old ($SD=11.40$), with the youngest at 18 years old, and oldest at 69 years old. Most participants (63%) reported gender as woman, followed by gender reported as man (32%), and gender nonconforming/fluid or transgender (5%). Participants reported racial identity as White/Caucasian (84%), Asian/Pacific Islander (8%), Black/African American (5%) or other (5%). A large majority of participants reported their nationality as the United States (91%). Educational attainment indicated that most participants had received a college degree (43%), most commonly a Bachelor's (20%) or Master's (14%) degree. Additionally, many participants reported educational level of "some college" (29%), while others reported having completed high school or trade/vocational school (18%). Most participants reported themselves to be working at some level (78%).

Table 5
Participant Demographic and Background Characteristics

Variable	<i>M</i>	<i>SD</i>	<i>Percentage</i>
Age	28.92	11.40	
Gender			32% Male 62% Female 5% Other

Race	84% White/Caucasian
	4% Black/African American
	8% Asian/Pacific Islander
	4% Other
Education Level	1% Some High School
	16% High School/GED
	2% Trade/Vocational School
	29% Some College
	2% Professional Certificate
	6% Associate's Degree
	20% Bachelor's Degree
	14% Master's Degree
	3% Doctoral Degree
Employment	17% Full Time
	17% Part Time
	1% Unemployed
	21% Student
	45% Student and Working
Nationality	91% United States
	8% Other

Note. n=133 for all demographic characteristics

Participant Political Characteristics

Table 6 shows a summary of participant political characteristics. For political ideology via liberal/conservative identification, 60% of participants reported themselves as very liberal to

somewhat liberal, neither liberal nor conservative (26%), and somewhat conservative to very conservative (14%).

Table 6.
Participant Background Political Characteristics

Variable	Mean	SD	Cronbach's Alpha
Liberal/Conservative			19% Very Liberal 30% Liberal 11% Somewhat Liberal 26% Neither 5% Somewhat Conservative 7% Conservative 2% Very Conservative
Social Dominance	6.18	1.64	.69
Engagement with Political Violence	3.05	1.05	.80
Trust in Institutions	3.89	1.05	.67
Institutional Goals	3.65	1.12	.69

Note. The scales for each measure were: Social Dominance (1 = Strongly Oppose, 4 = Neither Oppose nor Favor, 7 = Strongly Favor), Engagement with Political Violence (1 = Never, 2 = Rarely, 3 = Occasionally, 4 = Sometimes, 5 = Frequently, 6 = Usually, 7 = All the Time); Trust in Institutions (1 = Strongly Trust, 7 = Strongly Do not Trust), and Confidence in Institution Goals (1 = Strongly Confident, 7 Strongly not Confident).

For the Engaging with Political Violence measure, the internal consistency was very good, meaning it measured what it intended to measure. Participants reported, on average, that they viewed or thought about political violence. Specific items on the scale indicated that the participants reported occasional viewing of political memes online ($Mean = 3.02$, $SD = 1.55$), rare viewing of violent memes online ($Mean = 1.88$, $SD = 1.32$), occasional thinking about political violence ($Mean = 3.14$, $SD = 1.53$), and occasional thinking about violence in general

(*Mean* = 3.30, *SD* = 1.46). The participants reported that they rarely discuss political violence on social media (*Mean* = 1.95, *SD* = 1.23).

The internal consistency of the Social Dominance Orientation measure was good. The overall mean indicated that participants generally favored the idea of equality and non-dominance by the majority. Individual items were examined. The means scores by items were as follows: The majority should “get a say over the minority” (*Mean* = 4.51, *SD* = 2.32), “we should push for group equality” (*Mean* = 5.56, *SD* = 2.10), “in setting priorities, all groups should be considered” (*Mean* = 6.92, *SD* = 1.53), and that “superior groups should not dominate inferior groups” (*Mean* = 6.72, *SD* = 2.99).

The internal consistency of the Personal Trust of Institutions measure was low but adequate. Overall, moderate trust in institutions was reported. An examination of individual items indicated that the highest level of trust was placed in Business/Commercial Organizations (Means = 5.91, *SD* 1.37), followed by the Federal Government Institutions (*Mean* = 5.38 *SD* = 1.58) Non-Profit Organizations (*Mean* = 4.31, *SD* = 1.47), and lastly Academic Institutions (*Mean* = 3.95, *SD*=1.53).

The internal consistency of the Confidence in Institutions to Achieve their Goals measure was also low but adequate. Overall, moderate confidence in institutional goals was reported. The highest levels of confidence were reported for Federal/National Government Institutions (*Mean* = 4.56, *SD* = 1.72), Business/Commercial Organizations, Non-Profit/Nongovernmental agencies (*Mean* = 3.31, *SD* = 1.44), and Academic Institutions (*Mean* = 3.18, *SD* = 1.54).

Participant Social Media Platform Use

Almost all participants (99.5%, n=185) reported using some social media at least once or twice a year (See Table 7). YouTube was the most commonly used social media platform, followed by Instagram, TikTok, Snapchat, Facebook, Reddit , X/Twitter, and WhatsApp. When run with each independent platform, participants reported moderate engagement with social media overall (Mean=3.24, *SD*=.84).

Table 7.
Social Media Use

<i>Variable</i>	<i>M</i>	<i>SD</i>
YouTube	4.70	1.40
Instagram	4.19	1.80
TikTok	3.43	2.12
Facebook	3.35	1.61
SnapChat	3.26	2.18
Reddit	2.72	1.56
X/Twitter	1.92	1.24
WhatsApp	1.89	1.65

Perception of Risk Index (PRI) and Symbolic Political Violence Index (SPVI) Characteristics

The PRI and SPVI were developed for this study. The PRI contained six items and the SPVI contained five items. Participants completed the PRI and SPVI after viewing each meme. The internal consistencies of the PRI and SPVI were calculated after each meme viewing. The results are summarized in Table 8. As can be seen in the table, both measures demonstrated very good internal consistency across time and meme. This indicates that symbolic political violence is highly associated with perception of risk.

Table 8.

Internal Consistency of the Perceived Risk Index and Symbolic Political Violence Index

	WBS Meme	ALS Meme	MCV Meme	Mon Meme
SPVI	.83	.88	.83	.74
PRI	.92	.89	.93	.90

Note. WBS = White Boy Summer, MCV = McVeigh, ALS = All Lives Splatter, Mon = Monarch

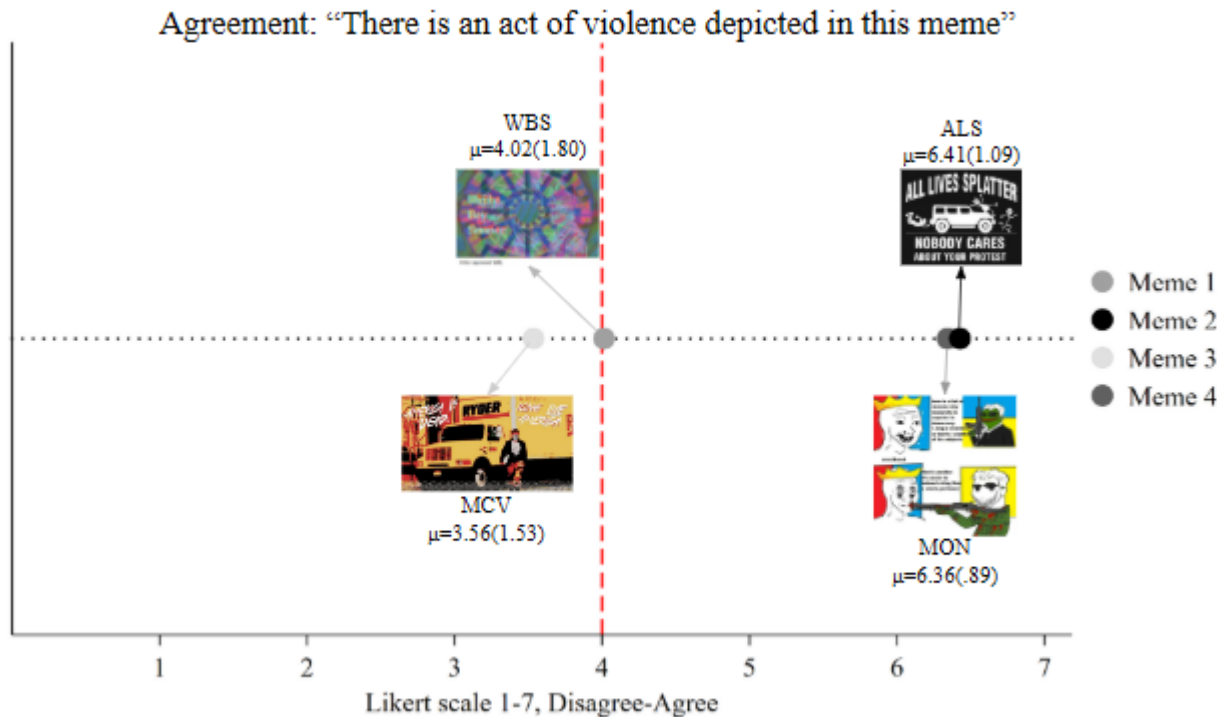
Meme Manipulation Check

One important question was to determine whether the memes were perceived as politically violent. After viewing each meme, the participants completed the SPVI. The 5 SPVI items were rated on a 1-7 scale (1 = strongly disagree, 4 = neither disagree or agree, 7 = strongly agree). For this study, the overall mean SPVI score was 5.23 ($SD = .75$). The mean SPVI score for each meme was also calculated. Results indicated that the ALS Meme was rated as most violent ($Mean = 6.16$, $SD = 1.07$), followed by the Mon Meme ($Mean = 5.94$, $SD = .87$), WBS Meme ($Mean = 5.00$, $SD = 1.35$), and the MCV Meme ($Mean = 3.85$, $SD = 1.19$). Taken together, these results indicated that the ALS, WBS, and MON Memes were clearly perceived as depicting violent content while the MCV Meme was viewed as less violent.

A closer inspection of the SPVI was conducted. SPVI item 1 question directly measured whether participants agreed that “There is an act of violence depicted in this meme. *Figure 5* shows the mean SPVI Item 1 score for each meme. The midline on the x-axis is “neither agree nor disagree.” The midpoint is an important threshold for these memes, as it indicates that participants qualitatively agree that the meme is violent. By selecting the agreement values greater than four, it can be determined that the memes are violent because the participants have declared it violent. If not, it would have to be taken at incremental value “approaching” violence for each meme. The ALS Meme and MON Meme clearly surpassed x-axis midpoint with an

average value exceeding 6 which represented “agree.” The WBS Meme slightly surpassed the midline of “neither agree nor disagree.” The MCV Meme did not surpass the midpoint. In summary, the ALS and Mon Memes were rated to be clearly depicting violence, the WBS meme was rated as slightly violent, and the MCV Meme was rated as not clearly depicting violence.

Figure 5
Symbolic Political Violence Index Item 1 Rating of Memes



SPVI item 2 asked participants to rate agreement with the following statement, “this meme is referencing real-world violent acts.” The results for this question followed a pattern similar to SPVI item 1: ALS Meme ($Mean = 6.21$, $SD = 1.28$), Mon Meme ($Mean = 5.37$, $SD = 1.38$), WBS Meme ($Mean = 5.69$, $SD = 1.52$), and MCV Meme ($Mean = 3.89$, $SD = 1.41$). Once

again, these results show that the ALS and MON Memes were considered as depicting real-world violence. Additionally, the WBS Meme was rated as depicting real-world violence. Finally, the MCV Meme was rated as not clearly depicting real-world violence.

SPVI Question 3 asked participants to rate agreement with the following statement, “this meme depicts lethal violence.” Ratings followed the same pattern as SPVI item 1: ALS Meme ($Mean = 6.25, SD = 1.20$), Mon Meme ($Mean = 5.93, SD = 1.32$), WBS Meme ($Mean = 4.11, SD = 1.79$), and MCV Meme ($Mean = 3.33, SD = 1.60$). An interesting finding on this item occurred with the WBS meme. The SPVI item 1 rating (asking about whether the WBS Meme depicted violence) was neutral. However, the participants were more likely to agree that the WBS Meme depicted lethal violence. Participants likely made a political interpretation and associated the image with symbols used by the Nazi regime, known for political violence.

SPVI Item 4 asked participants to rate agreement with the following statement, “this meme clearly presents a political viewpoint.” The results differed from the prior items because all meme ratings surpassed 4: ALS Meme ($Mean = 5.89, SD = 1.42$), Mon Meme ($Mean = 6.08, SD = 1.17$), WBS Meme ($Mean = 5.80, SD = 1.61$), and MCV Meme ($Mean = 4.85, SD = 1.59$). The Mon Meme was rated as depicting the most clear political viewpoint, followed by the Mon Meme, WBS Meme, and MCV Meme. These results indicate that all memes were sufficiently evocative of political meaning across participants.

SPVI Item 5 asked participants to rate agreement with whether the meme “clearly presents support for violence.” The pattern of results was similar to the other SPVI items assessing violence: ALS Meme ($Mean = 6.05, SD = 1.45$), Mon Meme ($Mean = 5.97, SD = 1.45$), WBS Meme ($Mean = 5.80, SD = 1.61$), and MCV Meme ($Mean = 3.78, SD = 1.57$).

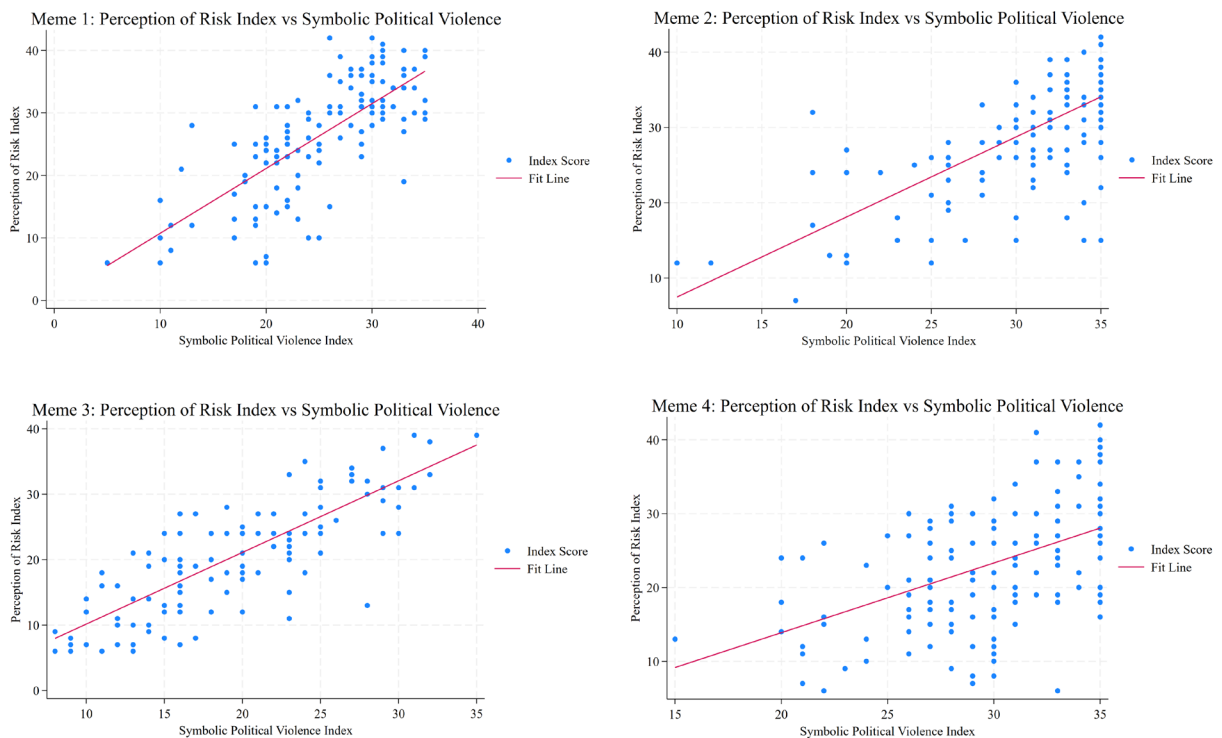
In summary, the participants agreed that the memes were political and violent. Additionally, the memes were largely viewed as violent, referencing real life violence, supportive of violence, and representing a political viewpoint. These results indicate that the experimental manipulation (i.e., exposure to violent memes) was successful.

Testing Hypothesis A: Correlation of PRI and SPVI

Hypothesis A was confirmed. Pearson's r was used to assess the relationships between SPVI and PRI for the memes. For the WBS Meme, $r(131) = .76, p < .001$. For the ALS Meme, $r(131) = .72, p < .001$. For the MCV Meme, $r(131) = .80, p < .001$. For the Mon Meme $r(131) = .52, p < .001$. Figure 2 depicts these relationships.

Figure 2

Scatterplots of Correlations between Perception of Risk Index and Symbolic Political Violence Index across Memes with Fit Line



Testing Hypothesis B: Social Media Use Predicting Perception of Risk

Correlations and Chi-Square analyses were conducted to determine if any of the demographic or background political characteristics were associated with social media use. Age, gender, race, education, social dominance orientation, trust in institutions, confidence in institutional goals, liberal/conservative, and engagement with political violence were evaluated for associations with overall social media use. Results indicated that age ($r(133) = -.34, p < .001$), race ($F(5, 127) = 3.05, p < .05$), nationality ($F(1,131) = 10.75, p < .01$), and engagement with political violence ($r(133) = .33, p < .001$) were associated with social media use. None of the other variables were associated with social media use.

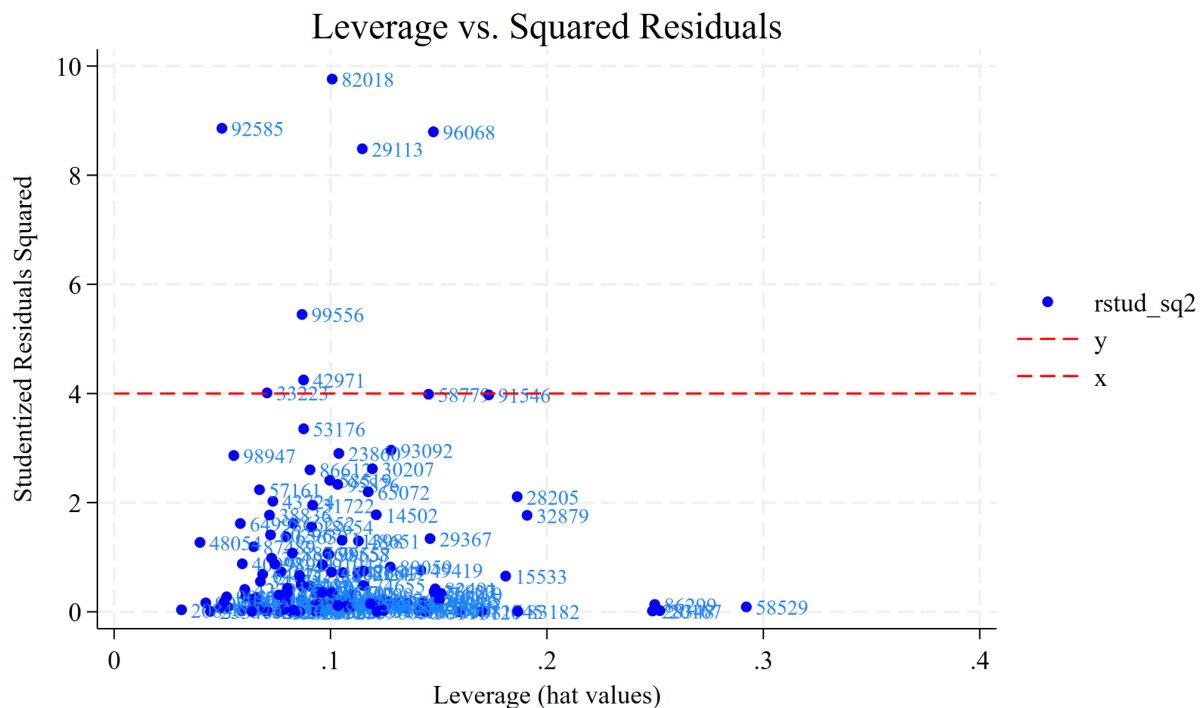
Hypothesis Testing Details

First, an ordinary least squares regression was used to determine the relationship between number of social media sites used and perception of risk. Age, race, nationality, engagement in political violence, and overall social media used were entered as predictors of overall PRI. The overall model was not significant [$R^2 = .046, F(5, 127) = 1.21, p = .31$].

Next, an ordinary least squares regression was used to evaluate the relationship between total social media usage and overall perception of risk. This model measured social media usage on all sites as an index score rather than a count of the number of sites. Results indicated that the overall model was not significant [$R^2 = .045, F(5, 127) = 1.21, p = .31$].

Post-estimation tests were conducted to ensure multiple regression assumptions were met. Variance inflation factor, Breusch–Pagan/Cook–Weisberg test for heteroskedasticity, Specification test, and Ramsey RESET Test for omitted variables were passed. These post-estimation tests ensure validity of the regression model. However, a Cook's Distance test showed

influential observations in the model. The latter was discovered using $D=4/n$ as a threshold, a value of approximately 0.03. Cook's D for this model showed a maximum value of 0.172, greater than the maximum threshold for influential observations. Plotting these influential observations as studentized residuals squared versus leverage, it could be seen that six observations are above this threshold (see Figure 7). Each observation represents one participant, as designated by their deidentified five-digit number.



was repeated, and the model was significant ($F(5, 127) = 1.37, p = .24$), but again social media usage was not a significant positive predictor of PRI ($p = .638$).

To further investigate this relationship, a new OLS regression was run using each of the social media sites as predictors. This model explained approximately 10% of the variance but the overall model was not significant ($R^2 = .10, F(11, 121) = 1.23, p = .28$), and X/Twitter was a significant negative predictor for total PRI ($\beta = -.77, p < .05$).

In sum, social media usage, when averaged across all social media, was not significantly associated with overall PRI. Most individual sites were not uniquely associated with overall PRI.

Testing Hypothesis C: Group Difference by Caption

Two groups of participants were formed through random assignment: those who viewed memes with an institutional caption; and those who viewed memes with a caption including only a “retrieved on” date. These groups are referred to as the caption group and the no caption group, respectively. All participants viewed that same four memes in random order. The hypothesis test required examination of the differences both between groups as well as within groups across the four memes.

Assuring that Random Assignment was Adequate

Prior to conducting the hypothesis test, the two groups were compared on relevant demographic variables and political background variables to assure that random assignment was adequate. Specifically, age, gender, race, education, liberal/conservative, social dominance orientation, trust in institutions, confidence in institutional goals, and engagement with political violence were compared between the caption and no-caption group participants. None of the between-group comparisons were significant. This means that the random assignment generated by Qualtrics XM was successful.

The test for this complex relationship required a 2x4 (two groups: caption/no caption by four memes: WBS Meme, ALS Meme, MCV Meme, Mon Meme) repeated-measures ANOVAs was conducted with PRI as the dependent variable. The justification for this test is the combination of measuring both the within-group variation as well as between-group variation, which required that participants be consistent, or show low variance in meme PRI scoring, which is important in post-estimation assurance.

The ANOVA main effect for group was not significant ($F(1, 131) = 1.00, p = .32$). The main effect for meme was large and significant ($F(3, 393) = 47.91, p < .0001$). This means that between memes, there were significant differences in how participants rated corresponding the PRI. The interaction between group and meme was not significant ($F(3, 393) = 1.08, p = .36$). This indicates that the caption had no significant effect on PRI. This effect was consistent when covariates were included. However, PRI significantly varied across the Memes. Post-hoc comparisons using the Bonferroni correction indicated that all memes were significantly different from each other. An examination of data indicates that the WBS and ALS memes had higher PRI scores than the MCV Meme and the Mon meme.

In a second repeated measures ANOVA the overall SPVI was used as the dependent variable. The ANOVA main effect for group (caption vs. no caption) was not significant ($F(1, 131) = .44, p = .51$). The main effect for meme was large and significant ($F(3, 393) = 153.07, p < .0001$). The interaction between group and meme was not significant ($F(3, 393) = .84, p = .47$). This indicates that the caption had no significant effect on SPVI. However, SPVI significantly varied across the Memes. Post-hoc comparisons using the Bonferroni correction indicated that all memes were significantly different from each other. An examination of data

indicates that the WBS, ALS, and Mon Memes had higher SPVI scores than the MCV Meme. These findings are consistent with manipulation check results.

Testing Hypothesis D: Comparison of Institutions

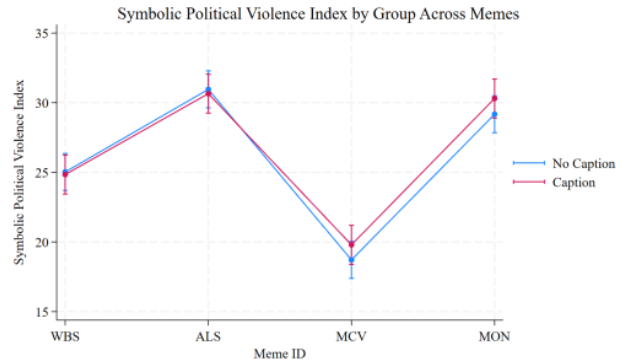
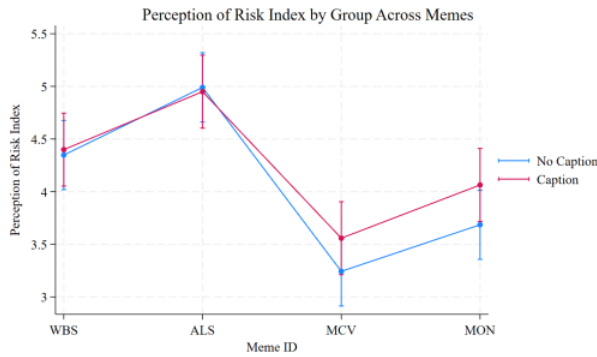
It was hypothesized that the between-group differences on PRI and SPVI would be smallest for the MON meme because it contained a non-professional institution in its caption. In order to assess each individual Meme on PRI, a mixed effects regression showed the main effect at each level on PRI. The reason for this test was to identify individual differences at each of these levels. The Hypothesis requires a comparison between the MON Meme and the others. Using the MON Meme as a baseline, the main effects follow WBS Meme ($\beta=3.66$, $p<.001$), ALS Meme ($\beta=5.88$, $p<.001$) and an no significant effect from MCV Meme ($\beta=-1.55$, $p=.08$).

To accommodate for individual differences, an additional mixed effects regression was performed using PRI and SPVI as the dependent variables illustrating the difference by meme. Using the MON Meme as a baseline for SPVI, the main effects follow: WBS Meme ($\beta=-4.1$, $p<.001$), ALS Meme ($\beta=1.87$, $p<.05$), and MCV Meme ($\beta=-10.56$, $p<.001$).

This suggests that WBS and ALS Memes elicited greater PRI than the MON Meme. Regarding SPVI, the WBS Meme and the MON Meme had a significantly lower effect on SPVI, and the ALS Meme had a significantly higher effect.

Figure 4

Mixed Effects Model of Main Effect of Each Meme on Perception of Risk Index, and Mixed Effects Model of Main Effect of Each Meme on Symbolic Political Violence Index



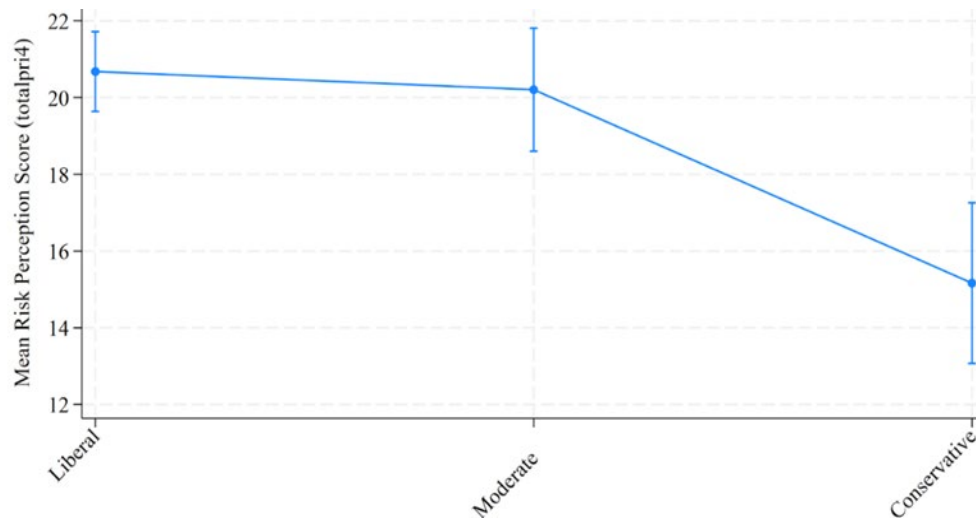
This hypothesis was further investigated using multiple T Tests at every level of PRI-per-Meme. No significant group differences were found at any level, meaning there was not even an independent group effect from one meme. Using SPV as the dependent variable, this was repeated to show no significant relationship between the group difference in the MON meme versus the remaining memes. Through these repeated statistical tests, a consistent null finding. This suggests that these constructs used to measure the group differences were not able to detect the hypothesized relationship.

Exploratory Analyses

An ordinary least squares (OLS) regression was used to evaluate the relationships between the perception of risk index and symbolic political violence index, and all other conceptually relevant study variables. These variables were: age, gender, race, nationality, social media usage, political ideology, engagement with political violence, social dominance orientation, personal trust in institutions, and confidence in institutions. Using these predictors, the model was overall significant ($R^2=.16$, $F(10, 122) = 2.31$, $p < .05$). Of the individual predictors, political ideology was the only significant predictor ($\beta=-0.90$, $p < .005$). This model was repeated using SPVI as the dependent variable. In this, the model was significant ($R^2=.24$, F

(10, 122) = 3.81, $p < .0005$). Of the individual predictors, political ideology was a significant predictor ($\beta = -0.88$, $p < .001$). A new grouped variable was created to pool liberal participants, moderate participants, and conservative participants *Figure 8* shows the effect of political ideology on perception of risk.

Figure 8
Perception of Risk by Grouped Political Ideology



Further Exploration of the Effect Captioning

Statistical testing on Hypothesis C showed that there was no significant difference between groups. This finding was examined using individual items as predictors for PRI. *Table 9* shows these differences. These changes were not powerful enough to achieve statistical significance.

In *Table 9*, the column headers show each meme, WBS, ALS, MCV, and MON. For each meme, there are three values listed: No Caption average, Caption average, and the percent change between the two. The rows, PR1 thru PR6 represent the six PRI items, with the questions listed below the table. In other words, this shows the full picture of Hypothesis C: the difference between the Caption and the No Caption groups. Reviewing the table shows that, despite no statistical significance in the overall model, there is variation between the two groups. The table is color-coded as follows: green means an increase in risk perception, and red means a decrease in risk perception. As *Table 9* shows, there were few instances of the same value of risk perception across the two groups.

Table 9

Itemized Perception of Risk Per Meme, With Percent Change Between Groups

	WBS Meme			ALS Meme			MCV Meme			MON Meme		
	No	Cap	%	No	Cap	%	No	Cap	%	No	Cap	%
PR1	3.23	3.33	+3.1%	4.1	4.09	-1.2%	2.69	3.06	+12.2%	2.40	2.56	+6.1%
PR2	4.67	4.62	-1.1%	5.3	5.48	+3.0%	3.11	3.67	+15.2%	3.57	4.14	+13.8%
PR3	4.50	4.49	-0.2%	5.0	5.00	-0.9%	3.29	3.56	+7.6%	3.89	4.29	+9.4%
PR4	4.40	4.40	-0.1%	4.9	5.00	+2.0%	3.31	3.51	+5.5%	3.57	4.10	+12.8%
PR5	5.01	4.95	-1.2%	5.5	5.40	-1.6%	3.57	3.89	+8.2%	4.62	4.83	+4.2%
PR6	4.27	4.60	+7.2%	5.0	4.73	-6.6%	3.49	3.67	+4.9%	4.06	4.48	+9.4%

Note. PR1(Personal Safety), PR2(Public Safety), PR3(Increased Violence), PR4(Cause Violence), PR5(Violent Attitude), PR6(No Change)

In the first row, PR1 can serve as an example. The question asked if each meme was a threat to personal safety. For the WBS meme, participants in the Caption group reported slightly more perception of risk than the No Caption group. The average response for the No Caption group was 3.23, while the average response for the Caption group was 3.33, or an increase of 3.1%.

Table 9 shows that the MCV and MON memes demonstrated a greater perception of risk in the caption group relative to the no caption group. When expanding the PRI to its six individual question items, this finding is consistent. This coordinates with individual T test findings, which showed group difference in these memes closest to statistical significance.

Key Takeaways

The clear takeaway from this study is that variation across the memes mattered most. Memes have a demonstrated relationship with risk perception and perceived violence, as well as political ideology. Where there was no relationship between treatment groups, the memes proved explanatory. This means that the constructs meant to measure group difference assisted in demonstrating the variation across the memes.

Conclusion

The hypothesis outcomes only represent a fraction of the story and must be interpreted to understand why they occurred. Though the majority of hypotheses could not be supported with the given data, there are still meaningful findings from these results.

Hypothesis A testing showed that the more violent the image, the more perception of risk was reported. This was an important finding, as these were both newly constructed measures, and worked as planned. This indicates that either of these measures are useful constructs that can be utilized in future research. At the same time, this confirms that participants did find the memes to be substantially violent. This helps confirm that the study indeed measured symbolic political violence, as reported by the participants.

Hypothesis B testing showed that social media did not have an impact on the perception of risk. Even with each social media site tested independently, there was no significant effect found. Although this result was not anticipated, it sheds light on the true sources of risk perception. This failure could be for several reasons. Conceptually, it may be that social media is not an actual agent of change, but merely a space where people import their preexisting worldview. It could also be that there are more complex elements to social media and risk perception than simply frequency of usage. Further investigation can elaborate on this effect.

Hypothesis C testing showed that the institution captions did not have a significant impact on perception of risk. The difference between groups was overall not significant, though there was a minor difference in the MCV and MON meme. Hypothesis D testing also showed that, since there was no significant group difference, there was no significant difference between institutions. In fact, the highest average change in perception of risk was found in the MON meme, which was anticipated to show the lowest change.

Together, these results show that the relationship between symbolic political violence, perception of risk, and institutions is not straight forward. It was anticipated that social media usage would amplify risk perception, and it did not. It was also anticipated that the institutions would amplify risk, and they did not. Furthermore, it was anticipated that Know Your Meme would amplify risk perception less than the other institutions, and it did not. However, a finding emerged that political ideology was a significant predictor for perception of risk.

There are important takeaways from all of these findings. The following chapter, Discussion, will systematically interpret and evaluate the findings herein. This includes a look toward existing research and knowledge which explains these results.

CHAPTER 5: DISCUSSION

Introduction

This chapter follows Chapter 4: Results, and aims to build upon these findings to make meaningful connections beyond this study. It begins with a brief summary of findings, and will highlight two key findings for deeper exploration. Next, the General Discussion faces three main questions about connection to research beyond this project. First, how do these findings relate to existing knowledge? Second, what do they mean within the broader field? Third, how might they advance and contribute to that knowledge? The final section of the chapter discusses the inherent limitations of the study before looking toward the Conclusion chapter.

Summary of Findings

This dissertation study aimed to investigate the relationship between symbolic political violence, institutions, and risk perception. Hypothesis A showed a positive correlation between two new constructs: a Perception of Risk Index (PRI) and a Symbolic Political Violence Index (SPVI). Hypothesis B testing showed that a greater use of social media did not correspond with greater perception of risk. Hypotheses C and D tested the experimental stimuli, the four image-macro memes. The test of Hypothesis C showed that the presence or absence of an institutional caption had no effect on risk perception, and the test of Hypothesis D showed no greater risk perception from memes captioned with professional institutions compared to the non-professional institution captioned meme.

Qualitative information was collected from the participants through a text-based question. The question stated, “How would you describe the defining characteristics of this meme? Please be as clear and specific as possible.” This question was meant to collect further details from the quantitative analysis and provide insight into the participants’ ideas. This provided a supplementary but ultimately more complete depiction of the participant risk perception experience, in the participants’ own words.

Key Findings

The chapter will focus on two key findings for deeper analysis. The first key finding is that institutions did not have a significant effect on risk perception. The result warrants further analysis as it represents an unexpected potential null result of a hypothesis. This means that the assumption that institutional captions *amplify* risk perception may be misguided or requires more research exploration.

The second key finding that will be explored for deeper analysis is that political ideology was a relevant factor in risk perception. Political ideology influenced risk perception more than any other demographic variable included in the study. Participants who identified as conservative rated the memes as less risky and less violent than liberal and independent participants.

Key Finding: No Effect of Institutions

A foundational assumption in this research study was that institutions would have an effect on risk perception. Testing showed no significant difference between participants who viewed memes with institutional captions versus participants who viewed memes with no captions. There could be many causes for this lack of effect (discussed further in the Limitations

subsection). For this analysis, however, the no-effect result will be taken at face value, meaning that institutional captions were not supported as a significant influence on risk perception.

In testing risk perception, the goal was to induce a change among participants during the experiment. The planned hypothesis regarding institutions presumed a directional result: that risk perception would *increase* with the presence of institutions. This hypothesis was borne out of attention to the Social Amplification of Risk Framework (Kasperson et al., 1988; Kasperson and Kasperson, 1996; Burns et al., 1990; Susarla, 2003). In this, risk signals like newspaper articles or government reports often amplify public risk perception. The study presumed that violent memes from institutions would yield greater risk perception due to the institutional influence alone.

The qualitative results demonstrated that participants could interpret the institutional caption as a risk signal. Participant 88259 (deidentified identification number) wrote about the MCV meme:

This meme depicts a cargo truck with a man walking forward, and the text "America is dead, Long live America" in reference to the saying about kings. I am unfamiliar with what the meme is trying to reference, although I can assume it is portraying a terrorist because of the source. (2025)

The source that the participant referenced reads: Joint Counterterrorism Assessment Team. (2022). "Use of Memes by Violent Extremists." National Counterterrorism Center, Department of Homeland Security, Federal Bureau of Investigation. Despite this qualitative acknowledgement, there are no responses directly linking the captions or institutions to the risk perception. Given that there was sufficient variance in participant risk perception, there must be

more influential factors in determining their assessment of each meme. Further discussion of this relationship and its potential implications can be found in the Limitations section below.

Key Finding: Political Ideology

It was expected that demographic variables would have an influence on perception of risk. Some of these expected influences included age, gender, trust for institutions, social dominance orientation and others. However, there was only one predictor which was observed consistently across models, and for every meme: political ideology.

Political Ideology was more predictive than: age; gender; education; race; employment; social media usage; how often one views memes online; how often one views violence online; how often one thinks about politics; how much one thinks about violence; and trust or confidence in institutions. There are several potential causes for this effect. In political psychology, it has been found that significant differences emerge in political ideology in reaction to negative stimuli (Hibbing et al., 2014). According to a meta-analysis by Jost et al., (2018), the strongest predictor for political conservatism was death anxiety. In connection with uncertainty in risk perception, the authors concluded that this relationship may ultimately “originate in psychological attempts to manage uncertainty and fear” (Jost et al., 2018, p.351).

Another plausible explanation for this effect is simpler: the memes feature mostly far right politics. Nazism, counter Black Lives Matter, the Oklahoma City Bombing, and Pepe the Frog may have been recognizable right wing political symbols for participants. Social identity has been linked with in-group bias (Tajfel and Turner, 1979; 2004). Considering the importance of political ideology as a predictor, this would indicate that in-group bias around violence was impacting risk perception. According to Ellemers et al. (2002), reactions to feelings of threat

depend on how committed a person is to their group social identity. In practice, this could explain why self-reported conservatives may find right wing violent memes less violent or risky due to their shared political viewpoint.

General Discussion

The study findings are meaningful for analysis and can be connected to the relevant literature underpinning the study as well as the broader field of political science. This subsection bridges the gap between this study and the body of academic research beyond. This portion begins with the existing and prior knowledge on the study topics, followed by applications to the broader field, and ends with the specific advancements and contributions which this study achieved.

Connections to Existing Knowledge

This study interrogated complex psychological, political, social, and philosophical questions like how violent memes impact perceptions of risk. The hypotheses were generated through the lens of social risk amplification. However, the results of this study demonstrated no significant amplification of risk. Therefore, the connection to the foundational literature should be recontextualized.

In the history of this research on symbolic violence and risk perception, there have been many approaches to finding connections. For example, violent media exposure and subsequent violent aggression are thought to have direct, causal, positive relationship (Anderson et al., 2003). This finding has been contested, however, with researchers pointing to other

environmental factors as equally relevant (Savage and Yancey, 2008; Ferguson and Kilburn, 2009).

In light of this found relationship between media and violence, some researchers sought out a causal mechanism to explain the relationship. Berkowitz and LePage coined the term ‘weapons effect’ to represent the way that people react to violent media. The term means that the presence of a weapon in media could make a person more aggressive. The original ‘weapons effect’ study measured fist-clenching speed when provided a negative stimulus (Berkowitz and LePage 1967). The researchers found that when participants viewed a picture of a weapon, as opposed to a neutral tool, their fist-clenching speed increased. This was once thought to be a premier measure of violence perception via arousal, but has been foundationally contested (Da Gloria et al., 1989; Benjamin Jr. et al., 2018).

Meme research is still relatively new in the social sciences compared to print and television media. One of the key findings from researchers is that memes tend to make violent messages appear innocuous through humor or obfuscation (Askanius and Keller, 2021). This process, sometimes called ‘memeification,’ can make research difficult. TikTok creator @zinaaweenaa described the memefication of far-right political groups. They noted that these groups can “become so satirical that, on the outside, we can barely understand how they’re communicating, what they’re saying, because so much of what they do is memefied” (@zinaaweenaa, 2025).

Demystifying and understanding the meaning of memes will require a diverse and dedicated coalition. There is no field or profession dedicated to this goal. The phenomenon of online media like memes, and their relationship to violence, are increasingly the focus of influencers, journalists, and researchers. As new evidence and findings are shared, social

scientists and other professionals will be able to confront potential causal mechanisms. This is a multidisciplinary phenomenon, and thus requires a coordinated effort across academic and professional domains. In the next section, the specific implications for the field of political science are discussed, followed by the specific advancements and contributions of this study.

Broader Field Considerations

The modern field of political science has been wrought with controversy (King and Langston, 2008), and upheaval (Monroe, 2015). In what MacKenzie (1971) called the political science of political science, there exists a tension between argumentation on one side, and tradition and authority on the other. In this tension, political science can be described as normative, both in the sense of describing and participating in various social norms, but also that the field “offers advice to rulers and citizens about their conduct” (p.289).

Risk perception research can and should be considered an integral part of political science. As discussed in Chapter Two, risk research is currently concentrated in business and actuarial, and medical fields. However, understanding risk as a relationship between a person and their physical, social, and spiritual environment can help offer insights into their political understandings. As well, political ideology is seen as a consistent predictor in even this rudimentary risk perception analysis.

Meme research specifically carries tremendous opportunity for political science as a field. Dean (2019) noted a “squeamishness” in political science around meme research. Because of this squeamishness, there is scant research available in the field. Dean also wrote, “I encourage political scientists to see socially mediated cultural production and exchange [specifically related to memes] not as some frivolous activity on the margins of politics, but as increasingly central to

how large numbers of predominantly young citizens experience politics” (2019, p. 256). This has important implications as to what could be missed in the field without targeted focus on memes, as this study has done.

Al Zidjaly (2017) noted that this lack of meme research in political movements missed important evidence that could provide insight for cross-cultural, linguistic, and theoretical coherence. Memes are a part of shaping global politics. Understanding global politics is a foundational component of political science as a field. The intentional inclusion of meme research into political science could be a significant achievement for the relevance of political science research. In order to offer advice to policymakers and publics efficiently and accurately, political science studies like this can make significant contribution. The next section discusses these contributions and significance.

Advancements and Contributions

Memes are an important vehicle for political communication. However, there is a challenge in understanding meme communication. This study is aimed at bridging the epistemic divide between the current state of academic research and the current state of the world regarding meme communication. Memes are a significant part of politics, and are primarily used by young people. People who were raised in the meme-laden internet culture are in their 20s and 30s. The impacts on politics, on violence, and on peace, will never be realized if researchers do not dedicate research to this culture. In the last decade, there have been numerous political mass murderers who specifically use memes in their manifestos and communications (Robins-Early, 2025). Political science has a history of responding to growing trends in political violence. For example, after the 1960s and 1970s, political science led a surge in terrorism studies

(Stampnitzky, 2013). Using lessons learned from the successes and failures of movements like this show the importance of intentional and structural investment into new phenomena.

This study was unique in many ways. First, it was the first to study risk perception using memes. Second, it was the first to study risk perception using measures associated exclusively with political violence. Third, this study was the first to specifically link viewing violent memes with risk perception. Fourth, this study was the first to create an index through which participants can quantitatively score the level of violence in a meme. Fifth, this was the first study to attempt a framing or priming effect on meme viewership.

With all of these unique contributions to the research community, there can be steps forward in using the understanding of memes to reduce human suffering from violence. These methodological and conceptual innovations enable researchers to understand how to use memes, manage meme usage, study meme proliferation, and capture meme access. Moreover, researchers can use these insights to improve large scale effects on risk perception, leading to more certainty, social trust, and feelings of safety.

Limitations

There are several limitations to this study, and each impacts the study differently. The first portion addresses sample size and population. The second section addresses improving the meme and caption as stimuli. The third discusses the PRI and SPVI constructs and suggests changes. The fourth section describes a problem in measurement on the MCV Meme, where participants did not know the meme and instead clustered at a neutral value. The fifth section addresses the contextual limitations and the research environment.

For a quantitative study, a larger sample size is always preferred. This would help add explanatory power to all analyses, and provide a clearer picture of the data. This could have been improved with greater understanding of the required statistical power and limitations of university information requests. The sample is also limited both geographically and by university affiliation. The study solely represents students at Kent State University who were enrolled in one or more courses at the time of the information request in Spring 2025. Simply revising the population, however, would only solve one part of the problem. Another part is that, even with a new population pool, participants may simply not know what the memes mean, or how to interpret them.

The meme viewing portion itself was limited in a number of ways. First, the number of memes was low. For example, the United States company Meta created a Hateful Memes Challenge dataset, which included over 10,000 memes (Kiela et al., 2020). This dissertation study was limited in the number of memes to select, as they were all legitimately selected as having been published by institutions. Increasing the number of memes could help glean clearer results and gain new insights from the memes. However, a large bank of memes is still only accessible to participants who are familiar with memes, leaving out proper measurement beyond memes.

Another key component missing from this model was the lack of a neutral control or even negative control. The Hateful Memes Challenge database includes many “false positives” or neutral and benign, non-hateful memes (Kiela et al., 2020). This means that some memes were not violent or a false positive of violence, while other memes were violent or a true positive. According to the participants, all four memes were at least somewhat violent. Including a not

violent selections of memes may help be able to isolate the violence in the meme for proper investigation.

The caption itself as a stimulus could be improved to benefit accuracy in determining difference between treatment and control groups. Since there is only evidence of one participant noticing the captions, it would make sense to enhance the effect of the caption. The goal is to increase or maximize the effect of the priming or framing of the institution, and associate it with the meme. The inherent limitation in the caption is that it is not an embodied experience. An embodied experience is when a human uses their sense to experience the world (Bengtsson, 2013). Participants did not visit an institution, meet an agent of an institution, nor have any experience of real-world interaction with the institutions used in the study. This necessarily means that the study did not capture a fully embodied institutional experience.

Among the variables, one particular item uncovered a methodological issue. For the PRI and SPVI variables for the MCV Meme, there was a clustering of scores around the middle value 4, meaning neither agree nor disagree. The likely cause of this clustering is that participants did not know what was in the meme. This is confirmed by the qualitative text response after the meme, where respondents specifically mentioned they do not know or do not understand the meme 21 times, and described it as confusing seven times. While some report that they did not understand, they still mentioned it being threatening or violent. One participant wrote, “I do not understand the underlying message of this meme. Even though I don't understand it, it feels violent to me” (Participant 90312, 2025).

There is also a contextual limitation to the research in that much work done in this space is completed using computer programs. In practice, this means the vast majority of work and innovations are conducted by multinational technology corporations and governmental bodies.

The Hateful Memes Challenge by Meta did not recruit social or behavioral scientists to examine their dataset, but rather machine learning and data scientists who produced computer program models. However, this infrastructure is less strong in political science. As these methods evolve in fields remote to political science, it puts a greater requirement on all interested fields to retain academic interactivity.

Conclusion

The lack of effect from institutions and the importance of political ideology has set the stage for discussion of the literature and field. Existing knowledge has done well to explain the foundations of risk, but is struggling to maintain explanatory power in the new environment of memes. Political Science, as a field, has much work to do in dedicated research toward memes and symbolic political violence. Though this study had some inherent limitations, it still offers valuable insights into the social world. The next chapter, Conclusion, will conclude analysis and look to the future of this study and other similar projects.

CHAPTER 6: CONCLUSION

Introduction

The future is being forged online. There are no indications that the relationship between humanity and cyberspace is dwindling. Politics, violence, risk perception, and social relations are dependent on this new space. Internet macro memes, one of the unique inventions of the online space, have been instrumental in developing this future. However, the impacts are largely unknown. Memes and the online space will continue to create new politics, new symbols, new risks, and new expressions of violence.

This study was conducted as the culmination of doctoral dissertation in political science. This study was proposed and defended as part of a prospectus to perform data collection and analysis. The aim of this study was to link the matters of symbolic political violence, institutions, and risk perception. The findings showed that these are related, complex, and require a great deal more investigation.

Returning to the Original Inquiry Background

This study was developed after years of experience as an amateur and professional political worker across the United States. I was able to crowdsource observations about the world through conversations with people from all walks of life. However, the more important background to this investigation is that I grew up online. I gained a sense of increasing online symbolic violence through phronesis, or practical wisdom, not formal investigation. I watched innocuous parts of society like vaporwave music and early web design transform into calling

cards for fascism (MacNair, 2023). I watched academics like Hans-Hermann Hoppe (depicted in the MON Meme) become cartoon symbols of violence (Ketcham, 2021). This transformation occurred through memes, online, and amongst primarily youth. This meme world is the future and the present, and even has the ability to reshape the past.

Potential Strategies to Mitigate Symbolic Political Violence

As it stands, the online space is still relatively unregulated (Bertola and Quintarelli, 2025). However, there are still several strategies being used by states, firms, and communities to moderate nefarious online activity. The current state of online regulation and safeguarding is a combination of state regulation and firm self-policing. Firms have hired human content moderators to sift through harmful online media, at great cost and controversy (Booth and Kimeu, 2024). States have begun strategies for a total social media ban for youth populations, which has already been instituted in Australia (Turnbull, 2025). Nongovernmental organizations can also lead lawsuits and legal actions against firms to demand changes (Bertola and Quintarelli, 2025). Researchers at National Taiwan University have built a tool called “Receler,” which acts as a supplement to AI models, preventing the models from generating harmful content in the first place (Lai, 2025). There have also been community-led efforts to reduce harmful online influences, such as The Tribunal, a system used by video game developer Riot Games, Inc. (League of Legends Wiki, 2020). This system was entirely user-run, including volunteer arbiters, mediators, and oversight roles.

It is clear that there will be many new attempts to curb the spread of online violence. These different methods have different origins and strategies, but are all welcome in the goal to reduce symbolic violence online. Creative and community-led strategies will continue to be

developed, and should be tested for efficacy by scholars and stakeholders. The shared goal must be to reduce harm and suffering from symbolic political violence. Key findings from this study, such as the importance of political ideology and the varied reactions to the memes rather than institutions, may serve as a basis for future inquiry designed to meet this goal.

Look to the Future

What I've learned from this study is that memes require participation; they are intersubjective. Without an audience to give them meaning, they are mere images. Armed with the findings from this study, I am able to see that people bring their individual differences to the meme experience. It was not the politics of the meme that influenced participants, it was their politics that influenced their experience of the meme. Neither memes nor social media are agents of change but people are.

There are several questions that arise from this study. Where does one look to understand this new meme world? What theories are available to explain this collective experience? What fields are best suited to address this phenomenon? What tools must be developed to attribute meaning to these changes? From a political science perspective, how does meme world organize power and politics? From a social science perspective, what challenges must be overcome to develop research projects in this increasingly memefied environment? These questions guided the initial inquiry, and remain central to my goals as a burgeoning scholar. With more studies like this, researchers, policymakers, cultural leaders, and moral entrepreneurs can help address the relationship between online violence, risk, and symbology.

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Appendix

Appendix 1.1: Flyer Recruitment

Political Violence Meme Research KSU Students Needed!



Study Title: Symbolic Political Violence and the Effects of Institutions on Perception of Risk

Investigators: Liam O'Brien, PhD Candidate, Department of Political Science; Tatsushi Arai, PhD, School of Peace and Conflict Studies

- Completion of survey enters you into a drawing to win one of four \$25 Amazon Gift Cards.
- Must be a Kent State student; 18 years or older

Appendix 1.2: Email Recruitment

Subject: Meme Research Study - KSU Students Needed!

Greetings,

My name is Liam O'Brien and I'm a PhD Candidate in the KSU Department of Political Science. I would like to invite you to participate in a research study!

The study: **Symbolic Political Violence and the Effects of Institutions on Risk Perception.**

You will be invited to take a short survey where you will answer questions and look at political memes. The survey takes about 15-20 minutes to complete. For completing the survey, you can enter a raffle to earn one of four \$25 Amazon gift cards.

To participate in this survey, please click the link below:

(_____ Long form Qualtrics link _____)

If you would like more information about this study, please contact me at lobrie10@kent.edu

Thank you very much!

Respectfully,
Liam O'Brien, PhD Candidate
Kent State University
lobrie10@kent.edu

This study has been approved by the Kent State University IRB (Study # 1386).

Appendix 2.1: Survey Flow

Show Block: Informed Consent (2 Questions)[Toggle Questions](#)[Add Below](#)[Move](#)[Duplicate](#)[Delete](#)

Then Branch If:
If Do you consent to participating in the study? **I do not consent** Is Selected [Edit](#) [Condition](#)

[Move](#)[Duplicate](#)[Options](#)[Collapse](#)[Delete](#)

End of Survey[Move](#)[Duplicate](#)[Customize](#)[Delete](#)

Set Embedded Data:
=
Add a New Field
[Add Below](#)[Move](#)[Duplicate](#)[Add From Contacts](#)[Options](#)[Delete](#)

Show Block: PRE: Demographic Characteristics (11 Questions)[Toggle Questions](#)[Add Below](#)[Move](#)[Duplicate](#)[Delete](#)

Then Branch If:
If Please select all of the following which are colors (you may select more than one):
Computer Is Selected [Edit](#) [Condition](#)
Or Please select all of the following which are colors (you may select more than one):
Vacuum Is Selected [Edit](#) [Condition](#)
Or Please select all of the following which are colors (you may select more than one):
Garage Sale Is Selected [Edit](#) [Condition](#)

[Move](#)[Duplicate](#)[Options](#)[Collapse](#)[Delete](#)

End of Survey[Move](#)[Duplicate](#)[Customize](#)[Delete](#)

Randomizer
Randomly present
of the following elements
Evenly Present Elements
Edit Count
[Add Below](#)[Move](#)[Duplicate](#)[Collapse](#)[Delete](#)

Group: MV1 IC
[Add Below](#)[Move](#)[Duplicate](#)[Collapse](#)[Delete](#)

Set Embedded Data:

```

graph TD
    A[Add a New Field] --> B[Show Block: Meme Viewing (1) (Institution Caption) (8 Questions)]
    B --> C[Group: MV0 NC]
    C --> D[Set Embedded Data:]
    D --> E[Add a New Field]
    E --> F[Show Block: Meme Viewing (0) (8 Questions)]
    F --> G[Show Block: POST: Trust (2 Questions)]
    G --> H[Show Block: Debrief (1 Question)]
    H --> I[Show Block: Raffle (1 Question)]
    I --> J[Then Branch If:]
    J --> K[End of Survey]
    K --> L[End of Survey]
  
```

The diagram illustrates a survey flowchart with the following components and actions:

- Add a New Field** (Green box)
- Show Block: Meme Viewing (1) (Institution Caption) (8 Questions)** (Grey box)
 - Actions: [Add Below](#) (Green), [Move](#) (Blue), [Duplicate](#) (Blue), [Add From Contacts](#) (Blue), [Options](#) (Blue), [Delete](#) (Red)
- Group: MV0 NC** (Blue box)
- Set Embedded Data:** (Green box)
- Add a New Field** (Green box)
- Show Block: Meme Viewing (0) (8 Questions)** (Grey box)
 - Actions: [Add Below](#) (Green), [Move](#) (Blue), [Duplicate](#) (Blue), [Add From Contacts](#) (Blue), [Options](#) (Blue), [Delete](#) (Red)
- Show Block: POST: Trust (2 Questions)** (Grey box)
 - Actions: [Add Below](#) (Green), [Move](#) (Blue), [Duplicate](#) (Blue), [Add From Contacts](#) (Blue), [Options](#) (Blue), [Delete](#) (Red)
- Show Block: Debrief (1 Question)** (Grey box)
 - Actions: [Add Below](#) (Green), [Move](#) (Blue), [Duplicate](#) (Blue), [Add From Contacts](#) (Blue), [Options](#) (Blue), [Delete](#) (Red)
- Show Block: Raffle (1 Question)** (Grey box)
 - Actions: [Add Below](#) (Green), [Move](#) (Blue), [Duplicate](#) (Blue), [Add From Contacts](#) (Blue), [Options](#) (Blue), [Delete](#) (Red)
- Then Branch If:** (Blue box)
 - Condition: **If All previous answers will remain anonymous and not be connected to any personal information share... I want to enter the anonymous raffle Is Selected** [Edit Condition](#) (Blue)
- End of Survey** (Dark Red box)
- End of Survey** (Dark Red box)
 - Actions: [Move](#) (Blue), [Duplicate](#) (Blue), [Options](#) (Blue), [Collapse](#) (Blue), [Delete](#) (Red)

Appendix 3.1: Survey Questionnaire

Symbolic Political Violence and the Effects of Institutions on Perception of Risk

Start of Block: Informed Consent

Q0 Informed Consent to Participate in a Research Study **Study Title:** Symbolic Political Violence and the Effect of Institutions on Risk Perception **Principal Investigator:** Tatsushi Arai, and Liam O'Brien You are being invited to participate in a research study. This consent form will provide you with information on the research project, what you will need to do, and the associated risks and benefits of the research. Your participation is voluntary. Please read this form carefully. It is important that you ask questions and fully understand the research in order to make an informed decision.

Purpose: This study is focused on understanding the effects of symbolic political violence and institutions on Kent State University students' perception of risk. The study aims to investigate several questions such as: How does viewing symbolic political violence affect perception of risk? If a meme is transmitted by a trusted institution, does this change perception of risk? To what extent does social media usage and thinking about political violence influence perception of risk from viewing symbolic political violence in memes?

Procedures: The study asks participants to answer questions in the following online survey. The survey begins with questions about your demographic background. Next, you will be prompted to view a selection of memes, viewing each for a period of 60 seconds minimum. Upon completing 60 seconds of viewing, you will be asked to answer questions about each meme. Afterward, you will be asked to voluntarily provide optional information for a follow-up interview about your answers. Completion of the survey should take no longer than 20 minutes.

Benefits: There are no direct benefits of participating in this study. However, your participation in this study will help scholars and practitioners to better develop tools and strategies for reducing the impacts of symbolic political violence online. This research can also help to build more scholarship in meme research, which is a constantly changing and interdisciplinary subfield of research that is very important to political expression.

Risks and Discomforts: There may be some questions or images (memes) which may make you feel uncomfortable. You will be shown four instances of memes displaying symbolic political violence, which some can find disturbing. If you do not wish to proceed, you may skip it and proceed to the next question or exit the survey. Your participation is expressly voluntary.

Confidentiality: No identifiable information will be collected. Your signed informed consent form will be separated from the study data and stored completely unlinked to your responses.

Future Research: Your de-identified information will not be used or shared with other researchers.

Compensation: Upon completion of the survey, you and all participants will be entered into a drawing to win one of four \$25 dollar gift cards to Amazon. The drawing will be held once data is collected from all participants. If you have won, you will be notified via email to collect your prize.

Voluntary: Your participation in this study is entirely voluntary. You may discontinue participation at any time without penalty or loss of benefits.

Age: Participation for this study is limited to individuals 18 years of age or older. By signing this form, you acknowledge and affirm that you are an appropriate age to participate in the study.

If you have any questions or concerns about this research, you may contact Tatsushi Arai at 330-672-2875. This project has been approved by the Kent State University Institutional Review Board. If you have any questions about your rights as a research participant or complaints about the research, you may call the IRB at 330-

672-2704. To participate click the button below. If you want to participate, please select "I consent." If you do not want to participate, please select "I do not consent," whereupon you will be removed from the survey.

Q1 Do you consent to participating in the study?

☐ I consent (1)

☐ I do not consent (2)

End of Block: Informed Consent

Start of Block: PRE: Demographic Characteristics



Q2 Please select all of the following which are colors (you may select more than one):

☐ Blue (1)

☐ Green (2)

☐ Computer (3)

☐ Garage Sale (4)

☐ Yellow (5)

☐ Vacuum (6)

Page Break



age The last most common type of multiple choice question is a drop down. You probably don't need much more about this one - it's a pretty typical way to give a lot of answers, like US states, without clogging up your survey or the screen. It'll work great on both desktop and mobile so you can rest easy knowing your respondents aren't too encumbered! Just like with multiple answer, you can select "Dropdown List" in the "Answers" section of the right-hand sidebar when you're editing a question.

Another quick note - our editor has some brainpower behind the fancy overlay. On this question, if you look in that right-hand sidebar, you can see that our choices are set with "Automatic Choices." When you write a question, if it follows a typical question pattern (e.g., "How much do you agree" or "Do you like" or things like that) our system will try to help out by suggesting some automatic answer choices. You can also manually choose what scale you want by clicking the dropdown arrow next to the automatic choices box.

How much do you agree with the following statement:

My tour experience has been off the charts!

▼ 18 (18) ... 99 (82)



gender How do you describe your gender identity?

☐

Man (1)

☐

Woman (2)

☐

Non-binary (3)

☐

Genderfluid (4)

☐

Transgender Woman (5)

☐

Transgender Man (6)

☐

Other (7) _____

☐

Prefer not to say (8)



nation What is your nationality?

☐ United States (1)

☐ Other (2) _____

race Choose one or more races that you consider yourself to be

☐ White or Caucasian (1)

☐ Black or African American (2)

☐ American Indian/Native American or Alaska Native (3)

☐ Asian (4)

☐ Native Hawaiian or Other Pacific Islander (5)

☐ Other (6)

☐ Prefer not to say (7)

Page Break



employment Mark all of the employment options that apply

- ☐ Full time (1)
- ☐ Part time (2)
- ☐ Unemployed (3)
- ☐ Retired (4)
- ☐ Student (5)
- ☐ Other (6)



education What is your highest level of education you've successfully achieved?

- ☐ Some High School (1)
- ☐ High School Diploma or GED (2)

- ☐ Trade or Vocational School (3)
- ☐ Some college (4)
- ☐ Professional Certificate (5)
- ☐ Associate's Degree (6)
- ☐ Bachelor's Degree (7)
- ☐ Master's Degree (8)
- ☐ Professional Degree (J.D, M.D, D.O, EdD) (9)
- ☐ Doctoral Degree (Ph.D) (10)



libcon What best describes your political ideology?

- ☐ Very liberal (1)
- ☐ Liberal (2)
- ☐ Somewhat liberal (3)
- ☐ Neither liberal nor conservative (4)

☐ Somewhat conservative (5)

☐ Conservative (6)

☐ Very conservative (7)

Page Break



socmed Social Media Usage- Please respond with the amount or frequency with which you use the following applications:

	Never (I have not once) (1)	Rarely (Once or twice per year) (2)	Occasionally (a few times per month) (3)	Sometimes (A few times per week) (4)	Frequently (A few times per day) (5)	Usually (Many times per day) (6)	All the time (Almost every hour of every day) (7)
Facebook (2)	☐	☐	☐	☐	☐	☐	☐

X/Twitter (3)	☐	☐	☐	☐	☐	☐	☐
Instagram (4)	☐	☐	☐	☐	☐	☐	☐
TikTok (5)	☐	☐	☐	☐	☐	☐	☐
Reddit (6)	☐	☐	☐	☐	☐	☐	☐
WhatsApp (7)	☐	☐	☐	☐	☐	☐	☐
Snapchat (8)	☐	☐	☐	☐	☐	☐	☐
YouTube (9)	☐	☐	☐	☐	☐	☐	☐



preconditions Please respond with the amount or frequency with which you do the following:

	Never (I have not once) (1)	Rarely (Once or twice per year) (2)	Occasionally (a few times per month) (3)	Sometimes (A few times per week) (4)	Frequently (A few times per day) (5)	Usually (Many times per day) (6)	All the time (Almost every hour of every day) (7)
View memes depicting political views online (1)	0	0	0	0	0	0	0
View memes depicting violence online (2)	0	0	0	0	0	0	0
Think about political violence (3)	0	0	0	0	0	0	0
Think about violence in general (4)	0	0	0	0	0	0	0

Discuss
political
violence
on social
media (5)

○ ○ ○ ○ ○ ○ ○

There are many kinds of groups in society: genders, ethnic and religious groups, nationalities, political factions. How much do you favor or oppose the ideas about groups in general?

Strongly oppose (1)	Oppose (4)	Somewhat oppose (5)	Neither oppose nor favor (6)	Somewhat favor (7)	Favor (8)	Strongly favor (9)
---------------------------	---------------	------------------------	--	-----------------------	--------------	-----------------------

The
majority
group
should get
a say over
the
minority
group (2)

○ ○ ○ ○ ○ ○ ○

We
should
not push
for group
equality
(1)

○ ○ ○ ○ ○ ○ ○

In setting
priorities,
we must
consider
all groups
(5)

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Superior
groups
should
not
dominate
inferior
ones (6)

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End of Block: PRE: Demographic Characteristics

Start of Block: Meme Viewing (1) (Institution Caption)

Q13 For the following section, you will be shown one of four memes. Each meme has been published by an institution to raise public awareness. The publishing institution is listed as the source. View each meme for a minimum of 60 seconds before proceeding to the next page.

Page Break

Q14  

Q15 Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)

Q16 Have you seen this meme before this survey?

☐ Yes (1)

☐ No (5)

Skip To: Q18 If Have you seen this meme before this survey? = No

Q17 If yes, where did you see this meme?

☐ X/Twitter (1)

☐ TikTok (11)

- ☐ Facebook (4)
- ☐ Instagram (5)
- ☐ Reddit (6)
- ☐ WhatsApp (8)
- ☐ Snapchat (9)
- ☐ YouTube (10)
- ☐ Other Website/App (Please write name below) (7)
-

☐ ☒ I have not seen this meme before this survey (2)

Page Break

Q18 How would you describe the defining characteristics of this meme? Please be as clear and specific as possible.



Q19 $\$ \{ \text{lm://Field/1} \}$

	Strongly agree (7)	Agree (6)	Somewhat agree (5)	Neither agree nor disagree (4)	Somewhat disagree (3)	Disagree (2)	Strongly disagree (1)
There is an act of violence depicted in this meme (1)	☐	☐	☐	☐	☐	☐	☐
This meme is referencing real-world violent acts (6)	☐	☐	☐	☐	☐	☐	☐

The violence depicted in this meme is lethal (3)

☐ ☐ ☐ ☐ ☐ ☐ ☐

This meme clearly presents a political viewpoint (4)

☐ ☐ ☐ ☐ ☐ ☐ ☐

This meme clearly presents support for violence (5)

☐ ☐ ☐ ☐ ☐ ☐ ☐



Q20 This next question type is probably one you're familiar with - the Matrix Table. There are a ton of different varieties here, but the most common and easiest to use is the "Single Answer" where each row (or Statement) needs to have one column (or Scale Point) selected.

Our system has a little trick up its sleeve to help out mobile respondents if the table is too big to fit on their device. As long as "Mobile Friendly" is checked on the right-hand sidebar, the system will automatically make matrix tables that are too big into our "Accordion" style, which changes the layout to resemble something more like a series of vertical multiple choice questions. Hey, lots of people take surveys on their phones now, and we don't want people quitting because the matrix table was too hard to decipher!

Please rate the following statements in terms of how much you agree.

	Strongly agree (7)	Agree (6)	Somewhat agree (5)	Neither agree nor disagree (4)	Somewhat disagree (3)	Disagree (2)	Strongly disagree (1)
This meme is a threat to my personal safety (7)	☐	☐	☐	☐	☐	☐	☐
There's flexibility in my ways to collect data (2)	☐	☐	☐	☐	☐	☐	☐
I can use a variety of question types to accomplish the goals I have (3)	☐	☐	☐	☐	☐	☐	☐
This meme is likely to cause violence in real life (4)	☐	☐	☐	☐	☐	☐	☐

This meme is likely to incite violent attitudes (5)

☐ ☐ ☐ ☐ ☐ ☐ ☐

This meme is **unlikely** to cause any change to violence or attitudes about violence (6)

☐ ☐ ☐ ☐ ☐ ☐ ☐

End of Block: Meme Viewing (1) (Institution Caption)

Start of Block: Meme Viewing (0)

Q21 For the following section, you will be shown different memes. Each meme is captioned with its retrieval date. You will be asked to answer questions after viewing each meme. View each meme for a minimum of 60 seconds before proceeding to the next page.

Page Break

Q22  

Q23 Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)

Q24 Have you seen this meme before this survey?

☐ Yes (1)

☐ No (5)

Skip To: Q26 If Have you seen this meme before this survey? = No

Q25 If yes, where did you see this meme?

☐ X/Twitter (1)

☐ TikTok (11)

- ☐ Facebook (4)
 - ☐ Instagram (5)
 - ☐ Reddit (6)
 - ☐ WhatsApp (8)
 - ☐ Snapchat (9)
 - ☐ YouTube (10)
 - ☐ Other Website/App (Please write name below) (7)
-

☐ ☒ I have not seen this meme before this survey (2)

Q26 How would you describe the defining characteristics of this meme? Please be as clear and specific as possible.



Q27 $\{\text{lm://Field/1}\}$

	Strongly agree (7)	Agree (6)	Somewhat agree (5)	Neither agree nor disagree (4)	Somewhat disagree (3)	Disagree (2)	Strongly disagree (1)
There is an act of violence depicted in this meme (1)	☐	☐	☐	☐	☐	☐	☐
This meme is referencing real-world violent acts (6)	☐	☐	☐	☐	☐	☐	☐
The violence depicted in this meme is lethal (3)	☐	☐	☐	☐	☐	☐	☐

This meme
clearly
presents a
political
viewpoint
(4)

☐ ☐ ☐ ☐ ☐ ☐ ☐

This meme
clearly
presents
support for
violence (5)

☐ ☐ ☐ ☐ ☐ ☐ ☐



Q28 This next question type is probably one you're familiar with - the Matrix Table. There are a ton of different varieties here, but the most common and easiest to use is the "Single Answer" where each row (or Statement) needs to have one column (or Scale Point) selected.

Our system has a little trick up its sleeve to help out mobile respondents if the table is too big to fit on their device. As long as "Mobile Friendly" is checked on the right-hand sidebar, the system will automatically make matrix tables that are too big into our "Accordion" style, which changes the layout to resemble something more like a series of vertical multiple choice questions. Hey, lots of people take surveys on their phones now, and we don't want people quitting because the matrix table was too hard to decipher!

Please rate the following statements in terms of how much you agree.

	Strongly agree (7)	Agree (6)	Somewhat agree (5)	Neither agree nor disagree (4)	Somewhat disagree (3)	Disagree (2)	Strongly disagree (1)
This meme is a threat to my personal safety (7)	☐	☐	☐	☐	☐	☐	☐
There's flexibility in my ways to collect data (2)	☐	☐	☐	☐	☐	☐	☐
I can use a variety of question types to accomplish the goals I have (3)	☐	☐	☐	☐	☐	☐	☐
This meme is likely to cause violence in real life (4)	☐	☐	☐	☐	☐	☐	☐
This meme is likely to incite violent attitudes (5)	☐	☐	☐	☐	☐	☐	☐

This meme is **unlikely** to cause any change to violence or attitudes about violence (6)

☐ ☐ ☐ ☐ ☐ ☐ ☐

End of Block: Meme Viewing (0)

Start of Block: POST: Trust

Q29 Choose how much you personally trust each of the institutions listed below

	Strongly trust (2)	Trust (3)	Somewhat trust (4)	Neither trust nor do not trust (5)	Somewhat do not trust (6)	Do not trust (7)	Strongly do not trust (8)
Federal/National Government Organizations (1)	☐	☐	☐	☐	☐	☐	☐
Non-Profit/Non-Governmental Organizations (2)	☐	☐	☐	☐	☐	☐	☐

Academic Organizations (3)	☐	☐	☐	☐	☐	☐	☐
Commercial Businesses (4)	☐	☐	☐	☐	☐	☐	☐

Q30 Choose how much confidence you have in each institution's ability to perform their organizational goals

	Strongly confident (1)	Confident (2)	Somewhat confident (3)	Neither confident nor not confident (4)	Somewhat not confident (5)	Not confident (6)	Strongly not confident (7)
Federal/National Government Organizations (1)	☐	☐	☐	☐	☐	☐	☐
Non-Profit/Non- Governmental Organizations (2)	☐	☐	☐	☐	☐	☐	☐

Academic Organizations (3)	O	O	O	O	O	O	O
Commercial Businesses (4)	O	O	O	O	O	O	O

End of Block: POST: Trust

Start of Block: Debrief

Q31 Debrief After completing surveys, a debrief is often considered an important part of a complete, ethical Informed Consent process (McCambridge et al., 2012; McShane et al., 2013; Verbeke et al., 2023). Though there was no deception or misconception, it is important to translate as much information as possible to study participants without influencing results. Part of this ethical strategy includes a participant's "right to know," and right to return to the pre-experiment condition. As such, this debrief is meant to serve to enhance understanding on the study design as well as study significance. This survey intends to measure changes in perception of risk based on the presence of institutions in the captions of memes. Each meme was published by a professional institution to raise awareness or provide information on political extremist memes: Federal/Governmental (NCTC, FBI, DHS), Nongovernmental (Anti-Defamation League: Center on Extremism), Academic (British Journal of Criminology), and Commercial Business (Know Your Meme). You, the participant, were placed randomly into one of two groups. One group was shown the memes with a citation identifying its publishing institution. The meme is shown with an attached caption saying where it was published. The other group was shown memes with just a "retrieved on" date, and no information about the publishing institution. All other survey questions remain the same between groups. The goal of this study is to measure differences in how participants respond to the questions before and after viewing the memes. The theoretical backing of this project stems from a psychometric research model called the Social Amplification of Risk Framework (Kasperson et al., 1988; 1990; 2022). Social risk perception, in practical terms, is often studied as a relationship between public perception and consequences of events (Lichtenstein et al., 1978; Cotgrove, 1982; Wynne, 1983, Fischhoff, Watson, and Hope, 1984, Kasperson et al, 1988). The SARF model describes that although real-world risk events occur at a specific likelihood and severity, the perception of that risk is subjective. Often, the institutions themselves have a major impact on perception of risk (Kasperson et al., 1988; Sovic, 1993; Frewer et al., 2003; Dai et al., 2022). The significance of this study lies in the selection of

symbolic political violence, in the form of memes, as the medium of risk communication. There are three main research questions posed by this project: 1) what is the relationship between a participant viewing symbolic political violence and perception of risk; 2) if a meme is published by a professional institution (governmental, nongovernmental, academic, commercial), what impact does this have on a participant's perception of risk; and 3) which factors can amplify or attenuate the change in perception of risk due to the presence of a professional institution alone. The findings of this study can help inform institutions, policymakers, researchers, and the public at large of the complex relationship between symbolic political violence and risk perception. Risk communicators may need to consider the amplification/attenuation effects of their particular institution when presenting technical risk assessments to the public. When people perceive risk, it can affect their worldview through cultivation (Gerbner and Gross, 1976; Shanahan and Morgan, 1999) leading to negative feelings (Van Loon, 2002) or even mobilizing for collective self-defense (Marcks and Pawelz, 2020). The overarching goal of this project is to aid in the reduction of human suffering and ending of violence. Symbolic political violence is already abundant, so the effects on perception of risk should be carefully monitored for possible social risk amplification. Using findings from projects like this can help risk communicating institutions develop an empirically based strategy to predict amplification effects of their messages. This way, institutions can satisfy their professional goals with a renewed understanding that their work can help reduce violence in all forms. Your participation in this study is an invaluable component in the development of a less violent AND more peaceful world.

☐ If you'd like to offer feedback of any kind to the researcher, please use the space provided (4)

End of Block: Debrief

Start of Block: Raffle

Q32 All previous answers will remain anonymous and not be connected to any personal information shared here. The winners will be randomly selected at the conclusion of the survey collection period. If selected, you will be contacted via email with the gift card. If you have any questions or concerns, please message Liam O'Brien at lobrie10@kent.edu

☐ I want to enter the anonymous raffle (4)

☐ I DO NOT want to enter the anonymous raffle (5)

End of Block: Raffle